

The Future of Growth and Human Labor Under Recursive AI Self-Improvement

Oliver Pardo

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Abstract

I develop an endogenous-growth model in which a monopolistic AI developer supplies AI production services and uses AI research services to improve AI efficiency, generating recursive self-improvement. I characterize the model's long-run equilibria under a finite upper bound on AI efficiency. The equilibrium regime depends jointly on the elasticity of substitution between labor and AI and on the level of this upper bound. When labor remains a production bottleneck, output per worker and wages grow at an exogenously given rate of labor-augmenting technological progress, and labor retains a positive income share. If the elasticity of substitution exceeds one and the upper bound is sufficiently high, the economy instead becomes AI-dominated. In this regime, wage growth exceeds the rate of labor-augmenting technological progress, but output per worker grows even faster. Labor's income share therefore converges to zero. With complementarity, long-run output-per-worker growth cannot exceed the rate of labor-augmenting technological progress, even without an upper bound on AI efficiency. With substitution, this growth rate increases without bound as the upper bound rises.

1 Introduction

Advances in artificial intelligence raise both hopes and fears. One hope is that increasingly capable AI could generate unprecedented economic prosperity. One fear is that workers who depend on labor income could lose relative economic power. This fear, however, does not require real wages to decline: workers may receive a smaller share of a much larger economy even as their wages rise. Can unprecedented prosperity coexist with a vanishing labor income share? Under what conditions could the world economy follow such a path? How long could it take before the transition became visible, and what economic signals would reveal it? What would happen to real wages and labor's income share along the way?

I study these questions by extending the Ramsey–Cass–Koopmans model. Competitive firms produce the final good with capital, effective labor, and AI production services. A monopolistic AI developer supplies these services and uses AI research services to improve AI efficiency, creating recursive self-improvement. Together, the elasticity of substitution between AI production services and labor and the maximum attainable level of AI efficiency determine whether human labor remains a long-run production bottleneck.

The escape from the bottleneck arises from two feedback loops. A technological loop runs from AI efficiency to AI research: greater efficiency produces more effective research services from a given

amount of research compute and thereby helps create still better AI. An economic loop runs from AI efficiency to output and back to AI research: better AI raises current production and expands the resources that can finance research compute and therefore further AI improvements. Research compute, however, competes with the inference compute used in final-good production.

I first characterize the labor-bottleneck regime. It arises when the elasticity of substitution between AI production services and labor is at most one, and can also arise when the elasticity exceeds one but the maximum attainable level of AI efficiency remains below an explicit threshold. Along equilibrium paths that converge to this regime, output per person and the real wage eventually grow at the exogenous rate of labor-augmenting technological progress, and labor's income share converges to a positive value.

The outcome changes when the elasticity of substitution between AI production services and labor exceeds one and the maximum attainable level of AI efficiency lies above the threshold. An AI-dominated long-run regime then becomes possible. Along equilibrium paths that converge to this regime, capital and effective AI production services eventually grow faster than effective labor. The long-run growth rates of output per person and the real wage exceed their labor-bottleneck values, and the interest rate is higher. Because output per person grows faster than the real wage, labor's income share converges to zero. AI-driven output growth raises wages despite the shrinking fraction of output paid to labor. Within the AI-dominated regime, long-run growth and the interest rate increase with the maximum attainable level of AI efficiency.

I also examine the role of the upper bound on AI efficiency. With complementarity, long-run output-per-person growth cannot exceed the exogenous rate of labor-augmenting technological progress, even without an upper bound. At unit elasticity and without an upper bound, I derive conditions for balanced growth in which output per person and real wages grow faster than this exogenous rate. With substitution, long-run growth and the interest rate in the AI-dominated equilibria increase without bound as the efficiency upper bound rises.

Simulations show that the AI-dominated long-run regime need not become visible quickly. The economy may remain on a moderate-growth path for a long time. Rapid growth of effective AI production services and an interest rate above the labor-bottleneck benchmark can appear before the output and wages start to accelerate. When the economy begins far from its steady state, the transition could also require a large initial research outlay relative to AI-industry revenue and temporary developer losses that are offset in present value by future monopoly rents. Alternative calibrations produce fast or slow transitions.

This paper has seven sections, including the present introduction. Section 2 relates the paper to the literature. Section 3 presents the model and defines equilibrium. Section 4 establishes sufficient conditions for equilibrium with a finite upper bound on AI efficiency and characterizes the associated long-run growth and distributional outcomes. Section 5 examines the three substitution regimes without an upper bound on AI efficiency. Section 6 presents some equilibrium numerical simulations. Section 7 concludes. The appendix contains the proofs and describes the algorithm used for the simulations.

2 Related literature

AI, growth, and automated research. The research technology builds on the endogenous-growth tradition, in which resources devoted to nonrival knowledge determine long-run growth (Romer, 1990; Aghion and Howitt, 1992; Jones, 1995). Recent work asks how AI affects the economy when AI can automate production, research, or both. Trammell and Korinek (2023) distinguish the automation of ordinary production from the automation of R&D. Mullainathan and Rambachan (2025) argue that AI may also reorganize the scientific process by bringing idea generation, theory selection, and other research activities that currently occur off screen into algorithmic workflows. The closest feedback architecture is Davidson et al. (2026): more capable AI makes subsequent AI research more productive, while higher output expands the resources that can finance that research. Korinek and Suh (2024) study how an exogenous automation frontier interacts with capital accumulation to shape output and wages during a possible transition to AGI. Korinek et al. (2026) take a complementary scenario-based approach. They map alternative paths of AI capabilities and adoption into GDP, factor shares, wages, occupational reallocation, and unemployment through 2030, but embed recursive improvement in the assumed path of AI task productivity. I instead place the technological and economic feedbacks inside a decentralized Ramsey economy in which consumption, saving, investment, compute, and prices are equilibrium outcomes. This also differs from quantitative scenario models such as Erdil et al. (2025), which combine AI development, automation, investment, and growth but abstract from the decentralized price system and market structure studied here.

Substitution, bottlenecks, and long-run growth. Following Davidson et al. (2026), I call an input or task a bottleneck when its slow improvement constrains aggregate progress even if other inputs or tasks improve rapidly. Aghion et al. (2019) apply Baumol’s cost-disease logic to AI and growth: if some essential production tasks remain difficult to automate, progress in automatable tasks need not translate into comparably rapid aggregate growth. Jones (2025) develops the corresponding mechanism inside R&D. The effect of AI on research depends jointly on the share of research tasks AI can perform, its productivity in those tasks, and the strength of complementarities across tasks. Tasks that AI cannot perform can therefore limit overall research progress. In my model, human labor no longer enters AI research, but it may remain essential in final-good production.

The production side connects this bottleneck logic to the classic result that sufficient substitution toward accumulable inputs can prevent their marginal product from vanishing and produce an asymptotically linear technology (Pitchford, 1960; Jones and Manuelli, 1990; Rebelo, 1991). Duffy and Papageorgiou (2000) emphasize that a high elasticity is not by itself sufficient: the productivity of the accumulable input must also be high enough. In the AI literature, Nordhaus (2021) identifies substitutability between information and conventional inputs as central to an economic singularity. Jones and Tonetti (2026) show instead how a small number of weak links can delay growth acceleration even when automation advances. Consistent with this mechanism, Demirer et al. (2026) find

that large AI-induced gains in coding activity attenuate sharply at the level of completed software projects and releases, and estimate strong complementarity between AI and human effort. Finally, Restrepo (2025) distinguishes work that remains a bottleneck from work that is no longer essential for growth. I obtain a related distinction without assuming that all work can be performed by AI. Gross substitution between AI services and labor is necessary but not sufficient for labor to cease being a bottleneck: attainable AI efficiency must also be high enough. I derive the threshold separating the labor-bottleneck and AI-dominated equilibria. The capability threshold in Gans (2025) concerns a different mechanism—whether AI directs research toward dense or exploratory regions of the knowledge space—rather than whether effective AI services can replace labor in final production.

Wages and the labor income share. The distributional results build on the task approach to automation. Acemoglu and Restrepo (2019) distinguish the displacement effect of automating existing tasks from two forces that can increase labor demand: the expansion of output generated by higher productivity and the creation of new tasks in which labor retains a comparative advantage, which they call the reinstatement effect. My model does not create new human tasks, so it does not contain this reinstatement margin. It captures the displacement and productivity forces. Labor and AI services enter a constant-elasticity-of-substitution (CES) production function. Holding capital and effective labor fixed, greater use of AI services raises labor’s income share when the elasticity is below one, leaves it unchanged when the elasticity equals one, and reduces it when the elasticity exceeds one. This mechanism is central to the analysis of the declining labor share in Karabarbounis and Neiman (2014). Recent work shows that automation can raise wages while reducing labor’s income share (Autor and Kausik, 2026). In the long run, Ray and Mookherjee (2022) show that capital accumulation can drive labor’s share to zero while real wages rise indefinitely, and Restrepo (2025) obtains a vanishing labor share when compute can perform all bottleneck work. The coexistence of rising real wages and a declining labor share is therefore not itself new. Farboodi et al. (2026) obtain a contrasting wage outcome: with endogenous capital accumulation, data-driven automation can generate explosive growth but stagnant long-run wages. My contribution is a sharper equilibrium characterization. Among the long-run regimes characterized here, real-wage growth exceeds the exogenous rate only in the AI-dominated regime, in which labor’s income share converges to zero. In every positive-labor-share regime, real-wage growth equals the exogenous rate. Within the AI-dominated regime, an exact expression links the wage-growth premium to the rate at which labor’s income share declines, while the AI-efficiency threshold determines which regime arises. The possible erosion of labor and human-consumption tax bases motivates the public-finance analysis in Korinek and Lockwood (2026). I do not study taxation, but the regime results identify when that policy problem can arise.

The aggregation of labor also matters for interpreting these results. Korinek et al. (2026) show that average wages can rise while wages for cognitive workers fall and their unemployment increases. My representative-worker model cannot characterize this heterogeneity within labor. It

instead isolates the conditions under which aggregate real wages rise faster than labor-augmenting technological progress while labor’s income share converges to zero.

Interest rates. AI-driven growth can also affect equilibrium returns. Faster expected consumption growth raises long-term real rates through consumption smoothing in [Chow et al. \(2026\)](#), and the AI scenarios in [Rachel \(2025\)](#) and [Melina and Villa \(2025\)](#) can raise the natural rate through different saving and investment channels. Using recent AI-sector investment, [Wachter and Wachter \(2026\)](#) infer scenarios with rapid output and AI-sector expansion and a higher risk-free rate. The sign is not universal in richer models. I study a narrower object: the net return on physical capital in a closed-economy Ramsey equilibrium. The return remains tied to labor-augmenting technological progress when labor is the bottleneck, but rises with attainable AI efficiency in the AI-dominated regime.

Measurement of AI services. The distinction between raw compute and the effective services it produces is informed by [Korinek and McKelvey \(2026\)](#), who distinguish physical computing capacity from quality-adjusted inference and R&D/training output and account for algorithmic progress. I retain separate compute flows for inference and research but summarize the technology that augments both in one AI-efficiency stock. [Coyle and Poquiz \(2025\)](#) document the broader difficulties of measuring AI investment and output, quality change, and changes in production processes. These measurement problems caution against interpreting the model’s AI-efficiency stock as a directly observed statistical series.

Market structure. The integrated developer is a tractable limiting case of the concentration forces studied by [Korinek and Vipra \(2025\)](#) and the AI market-power questions surveyed by [Gans \(2024\)](#). [Athey and Scott Morton \(2025\)](#) model AI as a priced intermediate input and show how upstream market power affects adoption, downstream prices, factor returns, and welfare. Evidence on model pricing, entry, and differentiation in [Demirer et al. \(2025\)](#) also cautions against treating current AI markets as a literal monopoly. I hold market structure fixed to isolate the dynamic mechanism. This accounting distinguishes the AI industry’s gross revenue from monopoly profit: inference and research expenditure are paid before the developer distributes the residual to its owner.

Contribution. The individual mechanisms studied here—automated research, production bottlenecks, declining labor shares, rising real wages, and changes in equilibrium returns—appear in the existing literature. I combine them in a decentralized endogenous-growth model with recursive AI improvement and a profit-maximizing AI developer. Across the long-run equilibria characterized here with a finite upper bound on AI efficiency, real wages grow faster than exogenous labor-augmenting technological progress if and only if labor’s income share converges to zero. Output per person then grows even faster, and an exact expression links the wage-growth premium to the

rate at which labor's income share declines. Within the AI-dominated regime, higher attainable AI efficiency raises both long-run growth and the net return on capital.

3 Model

Time is continuous. The economy contains a representative household, competitive final-good firms, and an integrated AI developer. Labor-augmenting technology A grows exogenously. AI efficiency B improves through autonomous AI research: the only current input used to raise B is research compute. The model therefore describes an economy in which the replacement of human researchers at the technological frontier has already occurred.

3.1 Population and labor-augmenting technology

Population grows at the exogenous rate $n \geq 0$:

$$\dot{N} = nN, \quad n \geq 0. \quad (1)$$

Each person supplies one unit of homogeneous labor inelastically. The aggregate labor endowment is therefore N . Because autonomous research uses no human labor, all labor is employed in final production and equilibrium will impose $L = N$.

The variable A measures the level of labor-augmenting technology. One unit of labor supplies A efficiency units, so effective labor is AL . Labor-augmenting technology evolves according to

$$\dot{A} = \gamma A, \quad \gamma > 0. \quad (2)$$

3.2 Household

The representative household owns physical capital and the AI developer. Let Π denote the net distribution the household receives from the AI developer. The household takes the paths of the net interest rate r , the wage w and Π as given. For a predetermined capital stock $K_0 > 0$, it chooses aggregate consumption C to solve

$$\max_{\{C_t\}_{t \geq 0}} \int_0^\infty e^{-\rho t} N_t \log(C_t/N_t) dt, \quad \rho > n, \quad (3)$$

subject to

$$\dot{K} = rK + wN + \Pi - C. \quad (4)$$

Admissible choices satisfy $C_t > 0$, $K_t \geq 0$, and $K(0) = K_0$. The variable r is the net real interest rate: depreciation has already been deducted from the rental return to capital. The household's Euler condition is

$$\frac{\dot{C}}{C} = n + r - \rho. \quad (5)$$

Its transversality condition is

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0. \quad (6)$$

3.3 Final-good production

Competitive firms produce the final good from physical capital K , effective labor AL , and effective AI production services X . Let $0 < \alpha < 1$ denote the capital elasticity, $\delta \geq 0$ the depreciation rate, and $\sigma > 0$ the elasticity of substitution between effective labor and effective AI production services. The CES weights ω_L and ω_X are strictly positive and sum to one. Production is

$$Y = K^\alpha Z^{1-\alpha}, \quad (7)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}. \quad (8)$$

Jobs comprise multiple tasks, and AI may substitute for human input in some tasks while complementing it in others ([Autor, 2013](#); [Acemoglu and Restrepo, 2019](#)). I abstract from differences across workers, occupations, and tasks by modeling homogeneous labor and aggregate effective AI production services. The elasticity of substitution governs the relationship between these aggregate inputs, rather than the allocation of individual tasks to humans or AI. This simplification focuses the analysis on aggregate wages and labor's income share, leaving differences in exposure and earnings across workers outside the model.

Let p_X denote the price of one unit of effective AI production services. Taking the input prices $r + \delta$, w , and p_X as given, a final-good firm solves

$$\max_{K, L, X \geq 0} \left\{ K^\alpha Z(AL, X)^{1-\alpha} - (r + \delta)K - wL - p_X X \right\}. \quad (9)$$

Define s_X as the elasticity of Z with respect to X :

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}. \quad (10)$$

The first-order conditions are

$$r = \alpha \frac{Y}{K} - \delta, \quad (11)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (12)$$

$$p_X = (1 - \alpha) s_X \frac{Y}{X}. \quad (13)$$

Constant returns and these conditions imply

$$Y = (r + \delta)K + wL + p_X X. \quad (14)$$

Solving the final-good firm's AI condition for its price yields the inverse demand schedule $p_X(X)$. The inverse absolute elasticity is

$$e_X \equiv -\frac{\partial \ln p_X(X)}{\partial \ln X} = \frac{1 - s_X}{\sigma} + \alpha s_X. \quad (15)$$

This demand schedule is an input to the developer's problem.

3.4 The AI developer

3.4.1 Effective AI production and research services

I use *compute* to denote the computational resources used to execute the numerical operations required by AI systems (Sastry et al., 2024). The variables U and M are rates of compute use rather than stocks of processors or data centers. I measure compute in units such that its price in final goods is one. Compute is therefore a rival, costly input, not a measure of AI performance.

I distinguish compute used to deploy AI from compute used to improve it. Inference compute U runs existing AI systems and produces outputs used in current final-good production. Research compute M is allocated to the development of better AI. It includes training and fine-tuning, but also computational experimentation, model evaluation, and other compute-intensive components of AI research. I therefore use the term *research compute* rather than the narrower term *training compute*. Korinek and McKelvey (2026) make a related distinction between compute used for inference and compute used for R&D and training.

Raw compute does not by itself determine the quantity of AI services produced. The variable B measures AI efficiency: AI services obtained per unit of compute. Thus U and M are raw computational inputs, while BU and BM are the corresponding flows of AI production and AI research services. This interpretation parallels measures of algorithmic efficiency based on the decline in compute required to achieve a given level of performance (Hernandez and Brown, 2020; Erdil and Besiroglu, 2022). A rise in B therefore represents an improvement in the algorithms that use compute, rather than an expansion of physical computing capacity. I assume that the same B augments compute in both activities.

Inference compute produces AI production services according to

$$X = BU. \quad (16)$$

Equation (16) maps inference compute into the flow $X = BU$ of AI production services. Research compute analogously produces BM , the flow of AI research services.

Figure 1 summarizes the production block.



Figure 1: Production of final output and AI production services. AI efficiency B augments inference compute U to produce $X = BU$, the flow of AI production services. These services combine with effective labor AL to produce Z , which combines with capital K to produce Y .

3.4.2 Autonomous improvement in AI efficiency

Given initial AI efficiency $B_0 > 0$ and an upper bound $\bar{B} \in (B_0, \infty]$, AI efficiency evolves according to

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad \psi(B) = 1 - \frac{B}{\bar{B}}, \quad \chi > 0, \quad 0 < \eta < 1. \quad (17)$$

Figure 2 summarizes the AI-efficiency improvement mechanism.

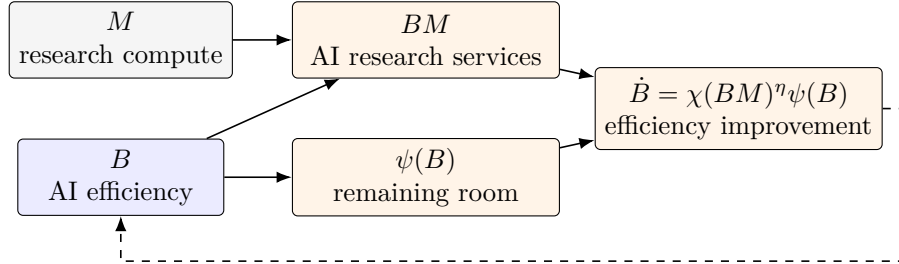


Figure 2: Autonomous improvement in AI efficiency. AI efficiency B augments research compute M to produce BM , the flow of AI research services. These services and the remaining room for improvement $\psi(B)$ determine $\dot{B}$, which increases B .

The parameter χ shifts research productivity. The parameter η governs returns to current research scale. The same AI-efficiency index B augments both inference and research compute. This common-productivity assumption is parsimonious but substantive. Because BM enters the law of motion for AI efficiency, a higher B makes each unit of research compute M more productive and creates recursive improvement. This is the AI-research feedback studied by Davidson et al. (2026). The term $\psi(B)$ reduces the remaining room for improvement as AI efficiency approaches $\bar{B}$. The uncapped specification is recovered at $\bar{B} = \infty$.

Evidence that ideas become harder to find motivates allowing research productivity to decline as technology advances. Bloom et al. (2020) document that research effort has risen substantially

while research productivity has fallen across several industries, products, and firms. Following this interpretation, Davidson et al. (2026) use their estimates to discipline diminishing returns in idea production. This evidence motivates diminishing returns, but it does not establish a literal finite technological ceiling. The term $1 - B/\bar{B}$ is a deliberately stronger reduced-form specification to make sure there is an equilibrium when AI and labor are substitutes.

The upper bound on AI efficiency $\bar{B}$ allows me to establish and compute equilibria for finite values of $\bar{B}$ and then ask whether they have a meaningful limit as $\bar{B} \rightarrow \infty$. Without the upper bound, recursive improvement may instead produce a finite-time singularity: one or more endogenous quantities—such as AI efficiency, output, capital, or research compute—become unbounded at a finite date, so the trajectory cannot be continued as a path of finite allocations.

3.4.3 Supply of effective AI production services

An integrated AI developer has exclusive ownership of B . It conducts research and sells effective AI production services. The developer sells X and purchases the $U = X/B$ units of inference compute needed to supply those services. Because X does not enter the law of motion for AI efficiency defined by Equation (17) directly, its within-date choice can be solved conditional on B, K , and L . Define optimized operating profit before research expenditure as

$$\pi(B) = \max_{X \geq 0} \left\{ p_X(X)X - \frac{X}{B} \right\}. \quad (18)$$

The notation $\pi(B)$ leaves its dependence on current K, A , and L implicit. The first-order condition equates marginal revenue to the compute cost of one additional service unit:

$$p_X(1 - e_X) = \frac{1}{B}. \quad (19)$$

Lemma 1 in Appendix A shows that problem (18) has a unique, strictly positive global maximizer characterized by condition (19). Thus, the developer charges the unit cost $1/B$ multiplied by the monopoly markup $1/(1 - e_X)$. The envelope theorem gives

$$\pi_B = \frac{X}{B^2} = \frac{U}{B}. \quad (20)$$

At a fixed X , a marginal increase in AI efficiency saves U/B units of the consumption good.

AI efficiency is nonrival in use. Once B has been developed, using it to augment one unit of inference or research compute does not reduce its ability to augment other units. This does not make effective AI services costless: BU and BM require the rival and costly compute inputs U and M . Nonrivalry also does not rule out excludability. An owner can restrict access to model weights, software, or AI services.

To see the resulting appropriability problem, suppose that competitive service providers had free access to B . Free entry would imply no operating rent. A developer that paid for M would improve a common AI-efficiency index B that competing providers could use without compensating

it. The developer would bear the research cost but could not appropriate the resulting benefits. Exclusive ownership allows the developer to capture part of the value created by a higher B and makes the return to research appropriable, as in the standard endogenous-growth treatment of nonrival ideas (Romer, 1990).

That being said, monopoly is an institutional assumption, not a technological implication of nonrivalry. Licensing, oligopoly, public funding, or open-source provision with research subsidies could also support investment in B . Each alternative would require an additional institution and would change prices, profits, and the distribution of rents. Recent work studies these alternatives in more detail (Korinek and Vipra, 2025; Gans, 2024; Demirer et al., 2025).

3.4.4 The dynamic problem

The AI developer uses M for research and owns B and any resulting improvements in AI efficiency. Taking the paths of r, K, A , and L as given and substituting the static solution to problem (18), the developer chooses the path of research compute M to increase AI efficiency B :

$$\max_{\{M_t\}_{t \geq 0}} \int_0^\infty \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt, \quad (21)$$

subject to (17) and B_0 given. Admissible research paths have $M_t \geq 0$ and finite total expenditure on every finite interval. The objective is the present value of net distributions, not the flow Π at a particular date.

Let q be the current-value shadow price of AI efficiency, measured in units of the current final good per unit of B . An interior research choice satisfies

$$q\chi\eta B(BM)^{\eta-1}\psi(B) = 1, \quad (22)$$

Equation (22) equates the marginal benefit of research compute to its unit cost. Holding current B fixed, an additional unit of M raises the rate of improvement in AI efficiency by $\chi\eta B(BM)^{\eta-1}\psi(B)$. Multiplication by q values the resulting efficiency gain, accounting for future profit gains and further improvements in AI efficiency.

The current-value costate equation is

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (23)$$

Rearranging gives the return equation

$$rq = \frac{U}{B} + q\eta \frac{\dot{B}}{B} + q\dot{B} \frac{\psi'(B)}{\psi(B)} + \dot{q}. \quad (24)$$

The first term on the right-hand side of Equation (24) is the inference-compute saving from greater AI efficiency. The second is the gain in research productivity. With a finite upper bound, the third

term is negative because a higher B leaves less room for further improvement. The final term is the capital gain on AI efficiency. Together these returns must equal the market return rq .

The developer's transversality condition is

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s \, ds \right\} q_t B_t = 0. \quad (25)$$

It rules out a path that leaves positive discounted value in the AI-efficiency state.

The preceding conditions are necessary for an interior optimum, but they do not by themselves establish global optimality. Lemma 2 in Appendix A gives a global verification result for the developer's research path. It uses a monotone transformation of the bounded AI-efficiency state and requires the maximized Hamiltonian to have a global supporting tangent at the candidate state. Global concavity is sufficient for this condition but is not necessary. The equilibrium existence proofs use a concavity test, whereas the quantitative analysis checks the weaker support condition directly whenever global concavity fails.

3.5 Equilibrium

I can now combine the conditions of the three agents. Capital-market clearing identifies household capital with the capital rented by final-good firms. Labor-market clearing gives $L = N$. AI-service market clearing identifies the developer's sales with the final sector's purchases. Combining the household budget, final-good zero profit, developer profit, and market clearing yields the aggregate resource constraint

$$C + \dot{K} + \delta K + U + M = Y. \quad (26)$$

Definition 1 (Equilibrium). *For exogenous initial values $A_0, N_0, K_0, B_0 > 0$, an equilibrium is an infinite-horizon trajectory*

$$\{K_t, B_t, C_t, q_t, L_t, U_t, M_t, X_t, Z_t, Y_t, r_t, w_t, p_{X,t}, s_{X,t}, e_{X,t}, \Pi_t\}_{t \geq 0} \quad (27)$$

such that for every date, the following conditions hold:

1. The intratemporal allocation and prices satisfy

$$L = N, \quad (28a)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad (28b)$$

$$Y = K^\alpha Z^{1-\alpha}, \quad (28c)$$

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}, \quad (28d)$$

$$e_X = \frac{1 - s_X}{\sigma} + \alpha s_X, \quad (28e)$$

$$r = \alpha \frac{Y}{K} - \delta, \quad (28f)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (28g)$$

$$p_X = (1 - \alpha) s_X \frac{Y}{X}, \quad (28h)$$

$$\frac{1}{B} = p_X (1 - e_X), \quad (28i)$$

$$X = BU, \quad (28j)$$

$$1 = q_X \eta B (BM)^{\eta-1} \psi(B), \quad (28k)$$

$$\Pi = p_X X - U - M. \quad (28l)$$

2. The four dynamic equations are

$$\dot{K} = Y - C - U - M - \delta K, \quad (29a)$$

$$\dot{B} = \chi (BM)^\eta \psi(B), \quad (29b)$$

$$\frac{\dot{C}}{C} = n + r - \rho, \quad (29c)$$

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (29d)$$

3. The two transversality conditions are

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0, \quad (30a)$$

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (30b)$$

The exogenous paths are $A_t = A_0 e^{\gamma t}$ and $N_t = N_0 e^{nt}$, and the predetermined stocks satisfy $K(0) = K_0$ and $B(0) = B_0$. At $\sigma = 1$, the CES equations use their continuous limits $Z = (AL)^{\omega_L} X^{\omega_X}$ and $s_X = \omega_X$. The trajectory is feasible for the agents' stated problems, the household and developer attain their optima, and final-good firms maximize profit. For a finite upper bound, $B_t < \bar{B}$ at every finite date.

Remark 1 (The economy without AI). As a separate boundary case, set $\omega_X = 0$ and $\omega_L = 1$, so that $Z = AL$. Effective AI production services have no productive use, and the developer has no revenue with which to reward improvements in AI efficiency. Thus $X = U = M = \Pi = 0$; B remains constant and does not affect allocations. With $L = N$, the model reduces to the Ramsey–Cass–Koopmans economy with logarithmic utility and exogenous labor-augmenting technological progress:

$$\begin{aligned} Y &= K^\alpha (AN)^{1-\alpha}, & \dot{K} &= Y - C - \delta K, \\ r &= \alpha Y/K - \delta, & \dot{C}/C &= n + r - \rho. \end{aligned}$$

The household transversality condition remains unchanged. The elasticity σ and the AI research parameters are irrelevant in this reduced economy; the interior AI price and costate conditions must not be imposed at $\omega_X = 0$. In particular, $wN/Y = 1 - \alpha$ and $(r + \delta)K/Y = \alpha$.

The system contains no separate household budget or final-good zero-profit equation because these are redundant after market clearing. The consolidated income decomposition follows from the equations in the definition:

$$1 = \frac{(r + \delta)K}{Y} + \frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y}. \quad (31)$$

It records how final output is divided among gross capital income, labor income, developer profit, inference compute, and research compute. It is an accounting consequence of equilibrium, not an additional equilibrium condition.

4 Equilibrium with a finite upper bound on AI efficiency

In this section, I characterize the long-run equilibrium behavior for various parameter configurations. The key finding is that long-run output-per-worker growth can exceed the rate of labor-augmenting technological change when the elasticity of substitution between AI production services and labor exceeds one and the upper bound on AI efficiency is sufficiently high.

To identify this threshold, I calculate the net return on capital implied by the production and pricing conditions as AI efficiency approaches its upper bound and labor's income share tends to zero. I compare this return with $\rho + \gamma$, the interest rate that sustains consumption-per-person growth at the exogenous rate γ . When AI and labor are substitutes and the former return exceeds the latter, the model supports equilibrium paths along which capital and AI production services grow faster than effective labor, even though AI efficiency converges to a finite limit.

4.1 The return to capital and the efficiency threshold

A finite upper bound restricts AI efficiency, but not the compute used to supply AI production services. To identify the return when labor's income share vanishes, fix $B > 0$ and consider $s_X \rightarrow 1$.

Since $wL/Y = (1 - \alpha)(1 - s_X)$, this is the zero-labor-share limit.

Combine the final-good firm's demand for AI services in equation (13) with the developer's pricing condition in equation (19). The result is:

$$\frac{X}{Y} = B(1 - \alpha)s_X(1 - e_X) \longrightarrow B(1 - \alpha)^2. \quad (32)$$

The limit uses $e_X \rightarrow \alpha$ from Equation (15). The CES aggregator and its elasticity in Equations (8) and (10) give

$$\frac{Z}{X} = \left(\frac{\omega_X}{s_X} \right)^{\sigma/(\sigma-1)} \longrightarrow \omega_X^{\sigma/(\sigma-1)}. \quad (33)$$

Dividing production by Y yields $1 = (K/Y)^\alpha (Z/Y)^{1-\alpha}$. Hence

$$\frac{Y}{K} = \left(\frac{Z}{Y} \right)^{(1-\alpha)/\alpha} \longrightarrow \left[(1 - \alpha)^2 B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (34)$$

Substituting equations (32) and (33) into equation (34), and the resulting ratio into $r = \alpha Y/K - \delta$, equation (11), defines the net return on capital when $s_X \rightarrow 1$ as a function of B :

$$\mathcal{R}(B) \equiv \alpha \left[(1 - \alpha)^2 B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha} - \delta. \quad (35)$$

This is a return at a particular limit of the production and pricing conditions, not necessarily an equilibrium interest rate. For $\sigma > 1$, $s_X \rightarrow 1$ means that AI production services become arbitrarily large relative to effective labor. For $\sigma < 1$, $s_X \rightarrow 1$ means that effective labor services become arbitrarily large relative to AI production services. Whether the equilibrium interest rate converges to $\mathcal{R}(\bar{B})$ depends on both σ and $\bar{B}$, as shown below.

By the Euler equation (5), consumption per person grows at $r - \rho$. A long-run equilibrium in which consumption per person grows at γ therefore requires $r = \rho + \gamma$. Define $\mathcal{B}(\sigma)$ as the AI-efficiency level at which the zero-labor-share return equals $\rho + \gamma$, the return required for consumption per person to grow at γ :

$$\begin{aligned} \mathcal{R}(\mathcal{B}(\sigma)) &= \rho + \gamma, \\ \mathcal{B}(\sigma) &= \frac{[(\rho + \gamma + \delta)/\alpha]^{\alpha/(1-\alpha)}}{(1 - \alpha)^2 \omega_X^{\sigma/(\sigma-1)}}. \end{aligned} \quad (36)$$

The inequalities between $\bar{B}$ and $\mathcal{B}(\sigma)$ —whether AI can attain sufficient efficiency— and between σ and 1—whether labor and AI are substitutes or complements—distinguish the long-run equilibrium regimes characterized below.

4.2 Equilibrium existence and long-run regimes

Define $g_{Y/N}$ as the growth rate of output per worker and g_w as the real-wage growth rate. The next proposition characterizes the long-run behavior of equilibrium trajectories.

Proposition 1 (Equilibrium existence and long-run characterization). *Fix $0 < \bar{B} < \infty$. For each parameter configuration covered by a case below, there exists a nonempty open set of initial conditions (A_0, N_0, K_0, B_0) such that every initial state in that set has an equilibrium trajectory with the following properties:*

(i) **Complementarity.** *If $0 < \sigma < 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (37)$$

(ii) **Unit elasticity.** *If $\sigma = 1$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad s_X = \omega_X \quad (38)$$

(iii) **Substitution with a labor bottleneck.** *If $\sigma > 1$ and $\bar{B} < \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (39)$$

(iv) **AI-dominated growth.** *If $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \mathcal{R}(\bar{B}) - \rho > \gamma, \quad (40)$$

$$g_w \rightarrow \gamma + \frac{\mathcal{R}(\bar{B}) - \rho - \gamma}{\sigma} > \gamma, \quad (41)$$

$$r \rightarrow \mathcal{R}(\bar{B}), \quad s_X \rightarrow 1.$$

Proof. See Appendix A, [Proof of Proposition 1](#). □

The omission of the case where $\sigma < 1$ and $\bar{B} < \mathcal{B}(\sigma)$ is deliberate as the upper bound $\bar{B}$ on AI efficiency is a device to avoid a singularity when $\sigma > 1$.

The proof first normalizes growing quantities and shrinking efficiency gaps. For each parameter configuration covered by the proposition, I conjecture stationary values of these normalized variables. I then construct trajectories from every sufficiently nearby initial state, deriving equilibrium trajectories and checking they converge to the stationary conjectures. Local uniqueness results are given in the proof. Existence from arbitrary initial states and global uniqueness are not established. These analytical results guide the algorithm's boundary conditions (Appendix B).

Remark 2 (Wages and AI revenue in the AI-dominated regime). In the AI-dominated equilibria of Proposition 1(iv), output per worker grows asymptotically faster than real wages, even though both grow faster than labor-augmenting technological progress:

$$\lim_{t \rightarrow \infty} g_{Y/N} > \lim_{t \rightarrow \infty} g_w > \gamma.$$

Labor's income share therefore converges to zero, while the AI industry's revenue as a share of output converges to $1 - \alpha$.

Figure 3 shows where the long-run regimes characterized in Proposition 1 lie relative to $\mathcal{B}(\sigma)$. For $0 < \sigma < 1$, the equilibria above the threshold have output-per-worker growth converging to γ and a positive limiting labor income share. For $\sigma > 1$, these same long-run outcomes occur below the threshold. Above it, the regime is AI-dominated: output per worker grows asymptotically faster than γ and labor's income share converges to zero.

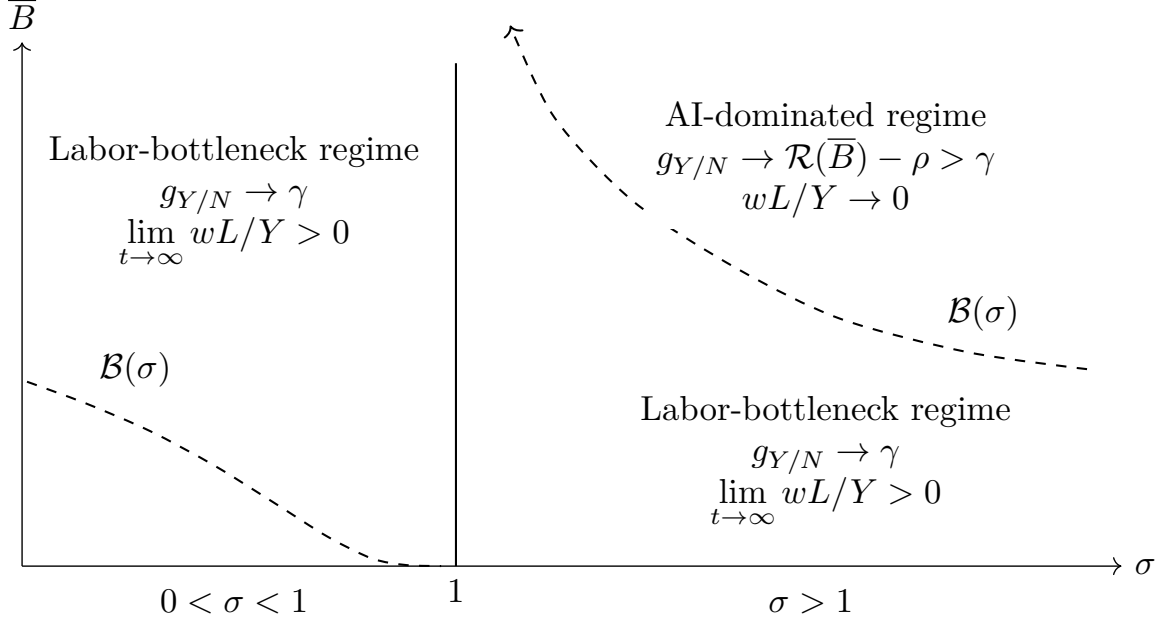


Figure 3: Constructed regimes with a finite upper bound. The vertical axis is linear and the curves are schematic rather than calibrated. For $0 < \sigma < 1$, $\bar{B} > \mathcal{B}(\sigma)$ is the condition for the positive-labor-share construction in Proposition 1(i). For $\sigma > 1$, the curve separates the labor-bottleneck and AI-dominated equilibria in Proposition 1(iii) and 1(iv).

Table 1 summarizes the four cases of Proposition 1. For $\sigma > 1$, an upper bound below $\mathcal{B}(\sigma)$ makes the return on capital at a vanishing labor share too low for capital and AI services to outgrow effective labor. Above the threshold, that return supports their faster growth along the AI-dominated equilibria. Across these equilibrium regimes, long-run real-wage growth exceeds γ if and only if labor's income share converges to zero.

Table 1: Long-run equilibrium characterization

Elasticity	AI efficiency bound	$\lim g_{Y/N}$	$\lim g_w$	$\lim wL/Y$	$\lim r$
$0 < \sigma < 1$	$\bar{B} > \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma = 1$	Any finite $\bar{B}$	γ	γ	$(1 - \alpha)\omega_L$	$\rho + \gamma$
$\sigma > 1$	$\bar{B} < \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma > 1$	$\bar{B} > \mathcal{B}(\sigma)$	$\mathcal{R}(\bar{B}) - \rho$	$\gamma + \frac{\mathcal{R}(\bar{B}) - \rho - \gamma}{\sigma}$	Zero	$\mathcal{R}(\bar{B})$

Section 5 studies the economy without an upper bound on AI efficiency. Those results require a separate equilibrium analysis.

5 Equilibrium without an upper bound on AI efficiency

I now remove the upper bound on AI efficiency and compare the three substitution regimes. The research law becomes

$$\dot{B} = \chi(BM)^\eta. \quad (42)$$

All other primitives remain as in Section 3. In Definition 1, I set $\psi(B) = 1$ and $\psi'(B) = 0$ and remove the upper restriction on B . Equilibrium still requires finite allocations at every finite date, global agent optimality, and both transversality conditions. Removing the upper bound does not by itself imply that B grows without bound.

The results differ in scope across substitution regimes. Under complementarity, I bound long-run output-per-worker growth by the rate of labor-augmenting technological progress. At unit elasticity, I give sufficient parameter restrictions for an analytically tractable balanced-growth equilibrium. Under substitution, I show how growth and interest rates increase with the efficiency upper bound and derive a return bound that applies at every date.

5.1 Complementarity

When $\sigma < 1$, effective labor remains necessary even if AI becomes arbitrarily efficient. Equation (8) gives an upper bound for Z and Y which does not depend on B :

$$Z \leq \omega_L^{\sigma/(\sigma-1)} AL, \quad (43)$$

$$Y \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} K^\alpha (AL)^{1-\alpha}. \quad (44)$$

Proposition 2 (Production bounds under complementarity). *Let $0 < \sigma < 1$ and $\bar{B} = \infty$. Along any feasible path satisfying the production equations (7) and (8), labor-market clearing $L = N$ in equation (28a), the uncapped research law (42), and the resource constraint (26), the following holds:*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \gamma. \quad (45)$$

Proof. See Appendix A, Proof of Proposition 2. □

Thus a permanent growth rate of output per person above γ is infeasible, although transitional growth can exceed it. Complementarity suffices to bound long-run growth, despite no bound on AI efficiency.

Proposition 3 (Long-run outcomes under complementarity without an AI efficiency upper bound). *Suppose $0 < \sigma < 1$, $\bar{B} = \infty$, and an equilibrium satisfies $B \rightarrow \infty$. Then*

$$\frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta} > 0, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0, \quad (46)$$

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad (47)$$

$$s_X \rightarrow \frac{1 - \sigma}{1 - \alpha\sigma}, \quad (48)$$

$$\frac{wL}{Y} \rightarrow \frac{\sigma(1 - \alpha)^2}{1 - \alpha\sigma} > 0, \quad \frac{p_X X}{Y} \rightarrow \frac{(1 - \alpha)(1 - \sigma)}{1 - \alpha\sigma} > 0. \quad (49)$$

Proof. See Appendix A, [Proof of Proposition 3](#). □

The capital and consumption ratios are conclusions, not assumptions. The proof first uses optimality and transversality to bound consumption and future research expenditure. It then shows that capital and consumption per unit of effective labor stay away from zero, while inference and research expenditure vanish relative to output. The remaining dynamics approach a Ramsey system, which yields convergence of $K/(AL)$ and $C/(AL)$ to positive constants. Convergence of consumption growth is therefore a conclusion, not an assumption. The proposition remains conditional on equilibrium existence and $B \rightarrow \infty$; it does not assume a BGP.

5.2 Unit elasticity

At unit elasticity, production becomes

$$Y = K^\alpha (AL)^{(1-\alpha)\omega_L} X^{(1-\alpha)\omega_X}. \quad (50)$$

Consider a balanced-growth path (BGP) with positive constant K/Y , C/Y , and M/Y . The AI-demand and monopoly conditions, equations (13) and (19), imply $U/Y = (1 - \alpha)^2 \omega_X^2$. Using $X = BU$ in production, and $g_M = g_Y$ in the research law, gives

$$g_Y = n + \gamma + \frac{\omega_X}{\omega_L} g_B, \quad (51)$$

$$(1 - \eta)g_B = \eta g_Y. \quad (52)$$

The first relation describes how AI improvement raises production growth; the second describes how expanding research resources sustain AI improvement. Solving these relations gives candidate growth rates. The next proposition establishes that they belong to an equilibrium.

Proposition 4 (An uncapped balanced-growth equilibrium). *Suppose $\sigma = 1$, $\bar{B} = \infty$, and*

$$0 < \eta < \alpha, \quad (1 - \alpha)\omega_X \leq \frac{1}{2}, \quad \eta + \omega_X < 1. \quad (53)$$

For every $A_0, N_0 > 0$, there exist $K_0, B_0 > 0$ and an equilibrium such that

$$g_{Y/N} = g_w = \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (54)$$

$$r = \rho + \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (55)$$

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad (56)$$

$$g_B = \frac{\eta(1 - \omega_X)}{1 - \eta - \omega_X}(n + \gamma). \quad (57)$$

Proof. See Appendix A, [Proof of Proposition 4](#). □

This is balanced growth, not a steady state in levels. The proposition constructs the initial stocks together with the equilibrium; it does not establish existence for arbitrary K_0, B_0 . Appendix C derives all allocation ratios and states sufficient conditions for equilibrium trajectories from nearby stocks that need not lie on the BGP. The condition $\eta + \omega_X < 1$ is necessary for this positive finite-rate BGP. The remaining restrictions are sufficient for the global optimality proof, not asserted to be necessary for equilibrium existence. The proof also covers $\eta > 1/2$: a change of state variable used only in the appendix permits a concavity argument without changing the research technology.

With any finite upper bound, the characterized unit-elastic equilibria have $g_{Y/N}, g_w \rightarrow \gamma$. Without an upper bound, continued AI improvement can sustain faster growth of both quantities while labor retains a positive income share. Thus the equivalence between a wage-growth premium and a vanishing labor share across the regimes with a finite upper bound does not extend to the uncapped economy. In the companion working paper *How Much Can Recursive AI Self-Improvement Raise Economic Growth?* ([Pardo, 2026](#)), I develop this unit-elastic benchmark as a framework for calibration and sensitivity analysis of long-run growth.

5.3 Substitution: growth and the AI-efficiency upper bound

With $\sigma > 1$, Proposition 1 establishes AI-dominated equilibria whose limiting interest rate and output-per-worker growth rate are $\mathcal{R}(\bar{B})$ and $\mathcal{R}(\bar{B}) - \rho$. For each fixed $\sigma > 1$, both increase without bound as $\bar{B} \rightarrow \infty$. This contrasts with Proposition 2: under complementarity, the long-run growth bound does not depend on an AI-efficiency upper bound.

The same function also bounds the interest rate at every date, with or without an efficiency upper bound:

$$r_t \geq \mathcal{R}(B_t), \quad \sigma > 1. \quad (58)$$

To see this, combine equations (32), (33), and (34) before taking the limit $s_X \rightarrow 1$:

$$\frac{Y}{K} = \left[B(1 - \alpha) \omega_X^{\sigma/(\sigma-1)} (1 - e_X) s_X^{-1/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (59)$$

For fixed B , this expression decreases with s_X and reaches its lower limit as $s_X \rightarrow 1$.¹ Substitution into $r = \alpha Y/K - \delta$ gives equation (58). Thus $\mathcal{R}(B)$ is the lowest return compatible with the production and monopoly conditions at that efficiency level, attained in the zero-labor-share limit.

If B grows without bound along a trajectory, this inequality requires $r \rightarrow \infty$ and $Y/K \rightarrow \infty$. Production also gives $Z/Y = (Y/K)^{\alpha/(1-\alpha)} \rightarrow \infty$, while the Euler equation (5) requires $g_{C/N} = r - \rho \rightarrow \infty$. These conditional implications and the comparison across finite upper bounds do not establish a finite-time singularity or settle equilibrium existence without an upper bound.

6 Quantitative equilibrium paths

6.1 Parameters and initial conditions

I compare four economies with the same parameters and predetermined initial stocks, varying only σ . Table 2 retains the macroeconomic benchmarks used in the earlier specification. The AI parameters are illustrative, not estimated. All four economies have autonomous research and a positive AI weight; the no-AI economy in Remark 1 is not one of these four scenarios.

¹Using $e_X = (1 - s_X)/\sigma + \alpha s_X$ from equation (15),

$$\frac{d \log[(1 - e_X) s_X^{-1/(\sigma-1)}]}{d \log s_X} = \frac{(\sigma - 2)(1 - \alpha\sigma)s_X - (\sigma - 1)}{\sigma(\sigma - 1)(1 - e_X)} < 0.$$

The denominator is positive for $\sigma > 1$. The numerator is affine in s_X and negative at both endpoints: $-(\sigma - 1)$ at zero and $-[1 - \alpha + \alpha(\sigma - 1)^2]$ at one.

Table 2: Parameters and scenario choices

Parameter	Value	Interpretation	Source or rationale
α	0.33	Capital elasticity	Standard one-third macro benchmark
δ	0.05	Annual capital depreciation	Annual benchmark; PWT comparison (Feenstra et al., 2025)
ρ	0.04	Annual household discount rate	Preference calibration
n	0.003	Annual population growth	Constant-rate approximation to the UN 2024–2100 world projection (United Nations, 2024)
γ	0.01	Annual growth of labor-augmenting technology	Illustrative long-run trend; see OECD (2025)
ω_X	0.20	Weight on effective AI production services	Scenario choice, not an observed industry share
ω_L	0.80	Weight on effective labor	$1 - \omega_X$
η	0.20	Research exponent	Illustrative choice; not estimated
χ	1.4378	Research productivity	Calibrated transition-speed parameter
σ	0.90; 1; 1.10; 1.50	AI–labor substitution	Four scenarios spanning the analytical regimes
$\bar{B}$	1.10 $\mathcal{B}(1.50)$ $\simeq 170.124$	Common upper bound on AI efficiency	Ten percent above the threshold for $\sigma = 1.50$

The constant n extrapolates a finite-period demographic approximation; it is not a claim that the UN projects constant population growth forever. The OECD benchmark for γ concerns measured labor-productivity growth, not A directly. I use it only as an approximate guide to the magnitude of long-run labor-augmenting technological progress. The parameter χ determines research speed relative to the other annual rates, so it is not merely a free time-unit normalization. I choose $\chi = 1.4378$ so that, in the $\sigma = 1.50$ equilibrium, the labor income share completes half of the distance between its date-zero value and its analytical limit after 50 model years. This is an illustrative timing normalization, not an estimate. Different values of χ produce faster or slower transitions and require the entire equilibrium path to be solved again. The horizontal axes below measure model years, not calendar forecasts. Likewise, the OECD’s discussion of AI-investment measurement supports treating ω_X as a scenario parameter, but does not identify its value ([Fonteneau et al., 2025](#); [OECD, 2026](#)).

The common upper bound is below $\mathcal{B}(1.10) \simeq 60.4$ million but above $\mathcal{B}(1.50) \simeq 154.658$. Thus $\sigma = 1.10$ retains a positive long-run labor share, whereas $\sigma = 1.50$ enters the AI-dominated regime. Changing the upper bound separately across scenarios would confound this comparison of elasticities.

The choice of initial conditions follows the question posed by each exercise. $A_0 = N_0 = 1$ fixes units. The capital stock K_0 and AI efficiency B_0 are predetermined states and are therefore specified before solving an equilibrium. Consumption C_0 and the developer’s shadow value q_0 are jump variables: I never import them from another economy or choose them to improve numerical

convergence. The BVP selects them anew for every reported path.

I set

$$A_0 = N_0 = 1, \quad K_0 = 4, \quad B_0 = 0.10\bar{B} \simeq 17.0124. \quad (60)$$

These common initial stocks place the economy closer to the capped $\sigma = 1$ terminal regime than the earlier uncapped reference: initial AI efficiency is 10 percent of its upper bound and initial capital per unit of effective labor is about 44 percent of its terminal $\sigma = 1$ value. I use these intermediate stocks so that the main figures display the transition without letting the financing needs of an extremely distant initial economy dominate the comparison. Holding the same two stocks across the four values of σ then isolates how the substitution elasticity changes the subsequent equilibrium path. These values are illustrative rather than estimated, and they are not a balanced-growth state of the capped economy. An exact constant-efficiency reference at $B_0 = \bar{B}$ would eliminate further improvements in AI efficiency and would lie outside the interior problem studied here. In each capped economy, the BVP selects C_0 and q_0 anew; neither jump variable is fixed at its uncapped reference value.

6.2 Numerical method and equilibrium admission

I retain the four-dimensional boundary-value architecture: two initial stock conditions and two terminal conditions selecting the stable equilibrium continuation. The appropriate regime with a finite upper bound from Section 4, not the uncapped unit-elastic BGP, determines the terminal conditions. Appendix B documents the continuation, equation reconstruction, and numerical checks.

In economic terms, the algorithm takes K_0 and B_0 as given and selects the initial consumption and shadow value that place the economy on the stable equilibrium path. It first constructs that path near the analytically characterized terminal regime and then moves it gradually to the prescribed initial stocks. Solving the boundary-value problem is only the construction step; the separate admission tests determine whether the resulting candidate is reported as an equilibrium.

A converged BVP is not sufficient for admission. I also check feasibility, independently reconstructed dynamic residuals, stability to longer horizons and tighter tolerances, both transversality conditions, and sufficient global developer optimality. The first three scenarios satisfy the global concavity test. For $\sigma = 1.50$, that stronger test fails at low counterfactual AI-efficiency levels, but the support condition in Appendix Lemma 2 is satisfied. Its analytical long-run bound is given in (82).

6.3 Growth, prices, and the distribution of output

Figures 4–6 present the transition in three steps: quantities and accumulation, prices and returns, and the distribution of output. Each panel includes the same four admitted trajectories. The three labor-bottleneck paths consequently appear close to one another in several panels. This compression reflects their common long-run regime; exact initial and limiting values of the central outcomes are

reported in Table 3.

The relative growth of effective AI production services is central to reading the figures. Rearranging (10) gives

$$\frac{s_X}{1-s_X} = \frac{\omega_X}{\omega_L} \left(\frac{X}{AL} \right)^{(\sigma-1)/\sigma}. \quad (61)$$

Differentiating this expression yields

$$\frac{d}{dt} \log \left(\frac{s_X}{1-s_X} \right) = \frac{\sigma-1}{\sigma} [g_X - (n+\gamma)]. \quad (62)$$

Thus the growth of $X/(AL)$ determines the direction and speed at which the AI share changes, conditional on σ .

Figure 4 first reports the growth rates of output, effective AI production services, and capital per unit of effective labor. These are $g_Y - (n+\gamma)$, $g_X - (n+\gamma)$, and $g_K - (n+\gamma)$, respectively. Growth rates show when the economy accelerates without compressing the large level differences that emerge across regimes.

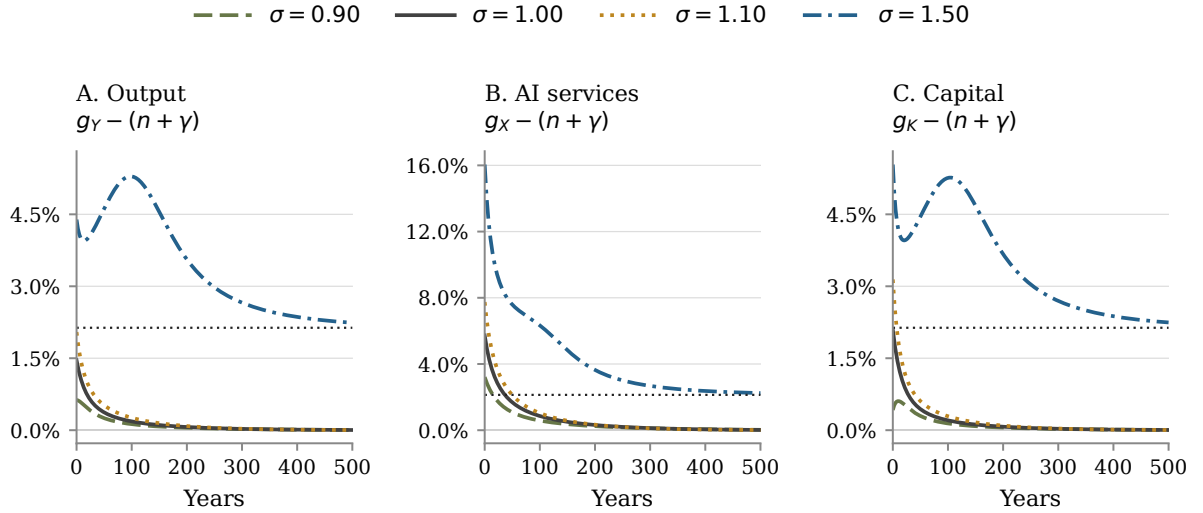


Figure 4: Growth of output, effective AI production services, and capital per unit of effective labor. Each panel compares the same four admitted numerical equilibrium trajectories over 500 model years. Panels A and C use the same linear percentage scale; Panel B uses a separate linear percentage range. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

Figure 5 next reports real-wage growth, the net interest rate, and the price of effective AI production services. The first two are rates; p_X is a relative-price level measured in units of the final good. Condition (19) implies $p_X = 1/[B(1 - e_X)]$. Because B approaches its finite upper bound and e_X approaches a value below one in every terminal regime, p_X converges to a positive finite level. It falls rather than grows without bound in all four simulations. I therefore report its level on a logarithmic scale, which keeps the four positive price paths visible despite their different magnitudes.

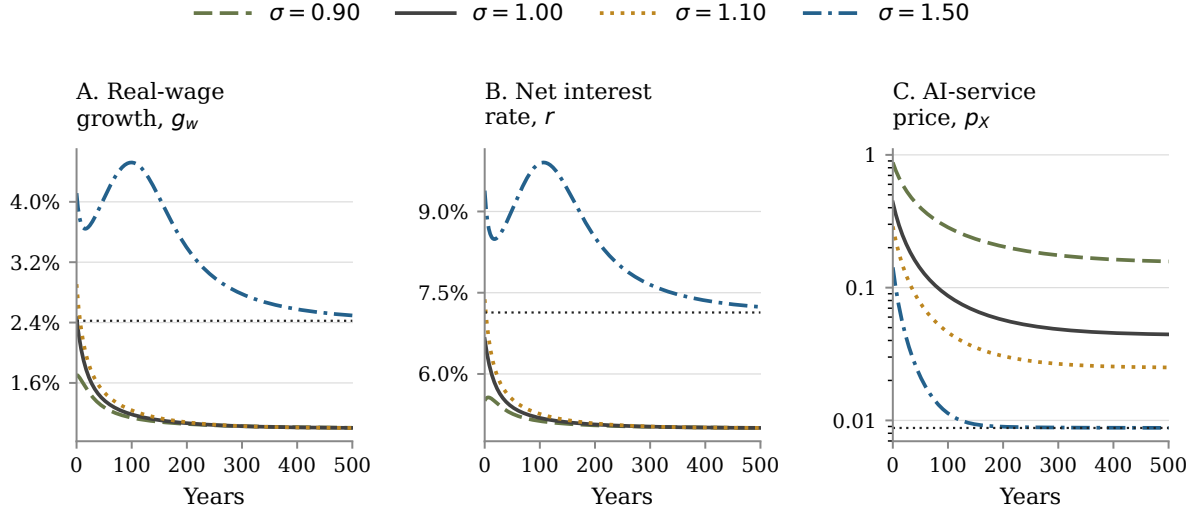


Figure 5: Wages, returns, and the AI-service price. Each panel compares the same four admitted numerical equilibrium trajectories over 500 model years. Real-wage growth and the net interest rate use linear percentage scales. The level of p_X uses a logarithmic scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

Finally, Figure 6 reports labor income and the three uses of AI-industry revenue as shares of final output. Combining final-good zero profit with $\Pi = p_X X - U - M$ gives

$$\frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y} = 1 - \alpha. \quad (63)$$

The four panels therefore exhaust the noncapital share of output. Gross capital income, $(r + \delta)K/Y = \alpha$, is constant and accounts for the remainder.

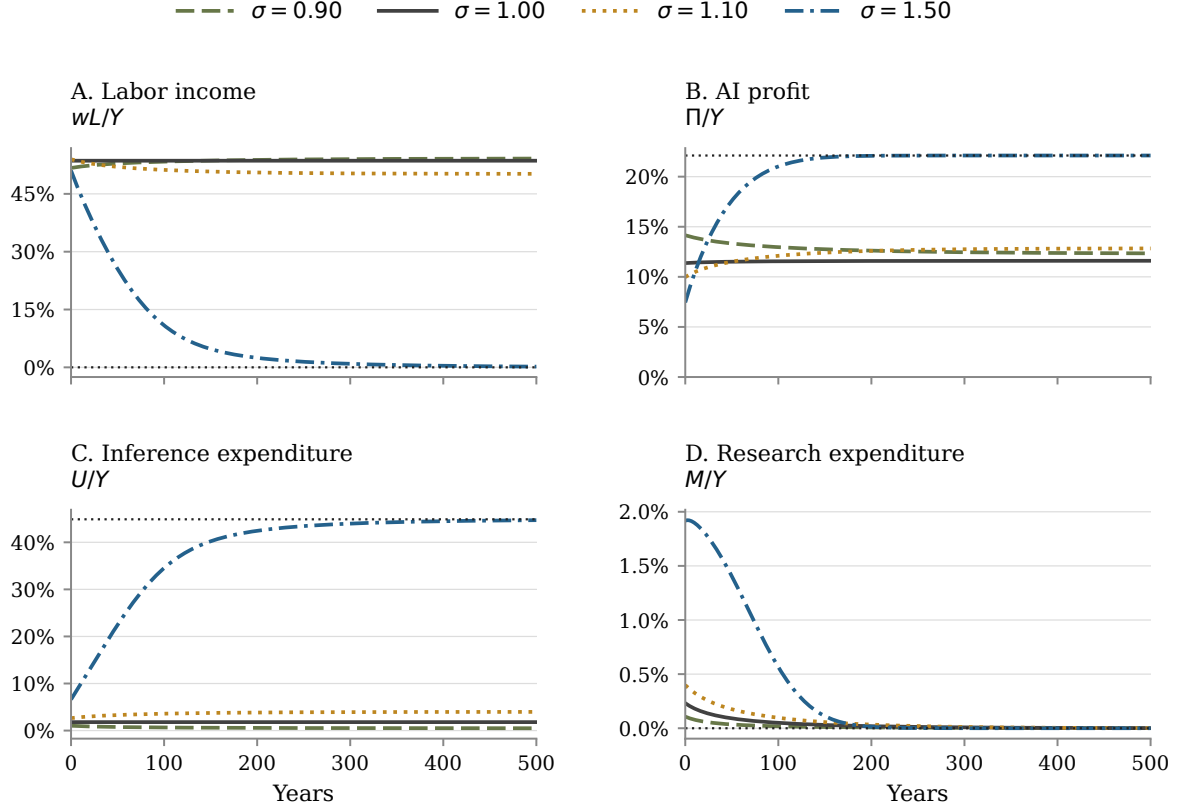


Figure 6: Distribution of the noncapital share of output. The panels report labor income, AI profit, inference expenditure, and research expenditure, each divided by final output. Together they sum to $1 - \alpha$; the omitted gross capital-income share is the constant α . Each panel compares the same four admitted numerical equilibrium trajectories over 500 model years and uses a linear percentage scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

Table 3: Initial and long-run values of the central outcomes

σ	Date	$g_Y - n$	g_w	r	wL/Y	$p_X X/Y$
0.90	0	1.629	1.711	5.508	51.7	15.3
	∞	1.000	1.000	5.000	54.2	12.8
1.00	0	2.458	2.458	6.646	53.6	13.4
	∞	1.000	1.000	5.000	53.6	13.4
1.10	0	3.046	2.909	7.380	54.0	13.0
	∞	1.000	1.000	5.000	50.2	16.8
1.50	0	5.390	4.115	9.385	51.0	16.0
	∞	3.135	2.423	7.135	0.0	67.0

Note: All entries are percentages. Date-zero entries come from the numerical equilibrium paths. Infinite-date entries are the analytical limits characterized in Section 4.

The first three scenarios approach per-person growth of 1% and a net interest rate of 5%. For $\sigma = 1.50$, their analytical limits are 3.135% and 7.135%, respectively. Real wages then grow asymptotically at 2.423%. Thus output per person grows 0.712 percentage points faster than the

real wage. By the accounting identity in equation (64) below, this implies an asymptotic decline of 0.712% per year in the log labor income share. The central distributional result is therefore not that real wages decline: workers become richer in absolute terms while their claim on total output vanishes. AI revenue meanwhile approaches 67% of output. These limiting rates describe the equilibrium continuations; they are not imposed as constant rates during the transition.

The AI-dominated path also displays signals before labor's income share has fallen sharply. At date zero, $X/(AL)$ grows at 16.051% and the net interest rate is 9.385%, well above the 5% labor-bottleneck benchmark, while labor still receives 51.0% of output. Moreover, because $L = N$, the accounting identity

$$\frac{d}{dt} \log \left(\frac{wL}{Y} \right) = g_w - (g_Y - n) \quad (64)$$

holds at every date. The gap between output-per-person and real-wage growth therefore does not merely predict a later distributional change: a positive gap is exactly the contemporaneous rate at which the log labor income share declines, and a widening gap means that decline is accelerating.

Equations (61) and (62) provide the structural explanation for the speed of the transition. A value $\sigma > 1$ makes the AI share rise with $X/(AL)$, but it does not make AI immediately replace labor. When σ is close to one, the exponent $(\sigma - 1)/\sigma$ is small and the share responds slowly even to large increases in relative AI services. At $\sigma = 1.50$, for example, the exponent is 1/3: increasing the odds $s_X/(1 - s_X)$ by a factor of ten requires a thousand-fold increase in $X/(AL)$. For elasticities closer to one, the required increase is larger still, conditional on the upper bound being high enough to support the AI-dominated regime.

The economy must also accumulate the inputs that produce this change. Neither K nor B can jump. Expanding X requires inference compute $U = X/B$, while improving B requires research compute M ; both uses compete with consumption and physical investment for current output. Moreover, $\eta = 0.20$ implies strong diminishing returns to a contemporaneous increase in research compute. As B rises, however, it reduces the compute required to supply a unit of X and makes subsequent research compute more productive. The feedback therefore strengthens gradually. Near the finite upper bound, $\psi(B)$ slows further efficiency improvement. The eventual acceleration in the AI-dominated equilibrium is consequently not an explosion of B : it is sustained by the expansion of capital and effective AI production services after labor ceases to be the binding input.

The date of takeoff is not a structural prediction of the model. Equation (61) implies an earlier transition when σ is farther above one or when X grows faster relative to AL . Within the model, greater research productivity χ , a higher initial B_0 , or a larger AI weight ω_X can advance the transition through different channels. Faster capital accumulation or slower growth of effective labor can also allow $X/(AL)$ to rise sooner. These changes are not interchangeable: σ and ω_X alter the production technology, B_0 changes the initial economy, and χ changes the research clock. The effects of changing η or $\bar{B}$ on the takeoff date do not have an unambiguous general-equilibrium sign because they also change research incentives and the terminal equilibrium. These comparative mechanisms indicate how the transition may move, but do not identify a date without recalculating

the equilibrium path.

The $\sigma = 1.50$ path displays this mechanism under the calibrated research speed. The labor share completes 10 percent of its decline after 8.3 model years, half after 50 years, and 90 percent after 145.2 years. At year 50, output per person grows at 5.616%, the real wage at 4.050%, and labor receives 25.5% of output. At year 100, the corresponding growth rates are 6.290% and 4.522%, while labor's share has fallen to 10.9%. By year 500, output-per-person growth is 3.239%, real-wage growth is 2.494%, and labor's share is 0.19%; all three are close to their analytical limits. The non-monotone growth response reflects the costly initial research surge, the subsequent acceleration in effective AI services, and eventual convergence as B approaches its upper bound. These dates illustrate an equilibrium under one research-speed calibration. They are not estimates of when an actual AI transition will occur.

Substitution alone does not produce this outcome. With the same upper bound and $\sigma = 1.10$, the labor share instead approaches 50.2%, compared with 53.6% at $\sigma = 1$ and 54.2% at $\sigma = 0.90$. The upper bound's position relative to the elasticity-specific threshold is essential to interpreting the comparison.

6.4 A transition from the no-AI steady state

As an additional transition experiment, suppose that the economy is initially on the no-AI Ramsey–Cass–Koopmans steady state described in Remark 1. Before date zero, $\omega_X = 0$ and $\omega_L = 1$. AI efficiency is then irrelevant for production, so I assign it a low latent value, $B_0/\bar{B} = 0.01$, solely to initialize the economy after AI production and research become available. The no-AI steady-state capital stock and the two predetermined date-zero stocks are

$$\begin{aligned} \left. \frac{K}{AN} \right|_{\text{RCK}} &= \left(\frac{\alpha}{\rho + \delta + \gamma} \right)^{1/(1-\alpha)} = 5.9416, \\ A_0 = N_0 &= 1, \quad K_0 = 5.9416, \quad B_0 = 0.01\bar{B} = 1.7012. \end{aligned} \tag{65}$$

The corresponding pre-change values are $Y/(AN) = 1.8005$ and $C/(AN) = 1.4262$. At date zero, ω_X rises to 0.20 and ω_L falls to 0.80. I hold the two stocks in (65) fixed across the four values of σ . The value of K_0 is therefore not an additional calibration: it is the exact no-AI steady-state capital stock implied by the maintained macro parameters. The one-percent value for $B_0/\bar{B}$ is an illustrative low starting point, chosen to leave substantial room for AI improvement after the technology switch; it cannot be inferred from the pre-change economy because B has no economic role when $\omega_X = 0$. Keeping these two predetermined stocks common across σ isolates the role of substitution conditional on the same Ramsey capital stock and the same latent initial AI efficiency. Consumption can jump after the change in technology, and the developer's shadow value has no no-AI counterpart. The BVP therefore selects C_0 and q_0 anew in each positive-AI economy.

Figures 7–9 use the same variables, normalizations, scenario order, and scale conventions as the main comparison. All four paths are independently solved and pass the equilibrium-admission tests.

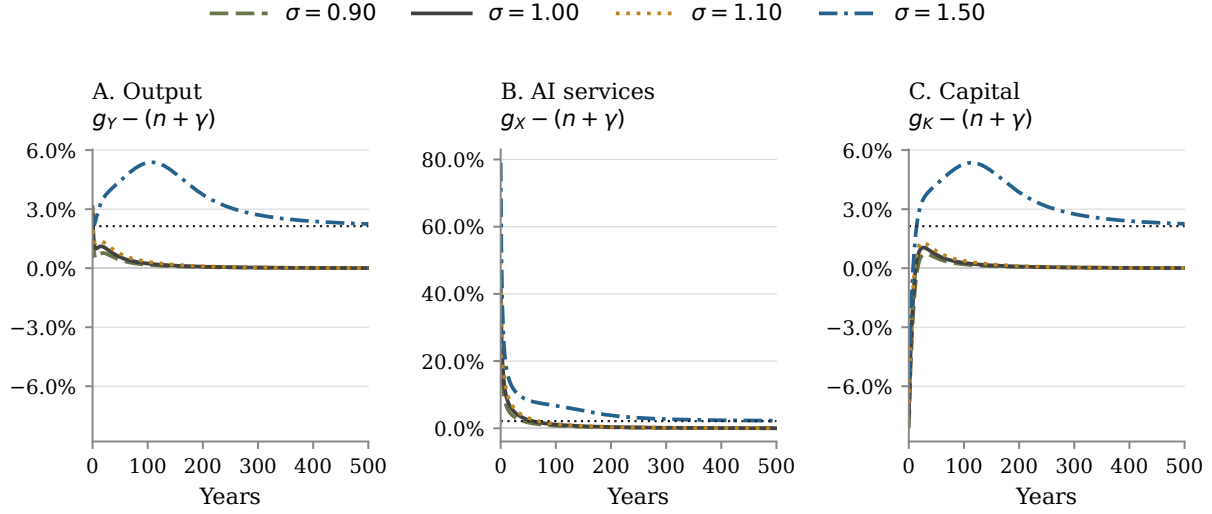


Figure 7: Growth of output, effective AI production services, and capital per unit of effective labor after the switch from the no-AI steady-state stocks. Each panel compares four admitted numerical equilibrium trajectories over 500 model years. Panels A and C use the same linear percentage scale; Panel B uses a separate linear percentage range. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

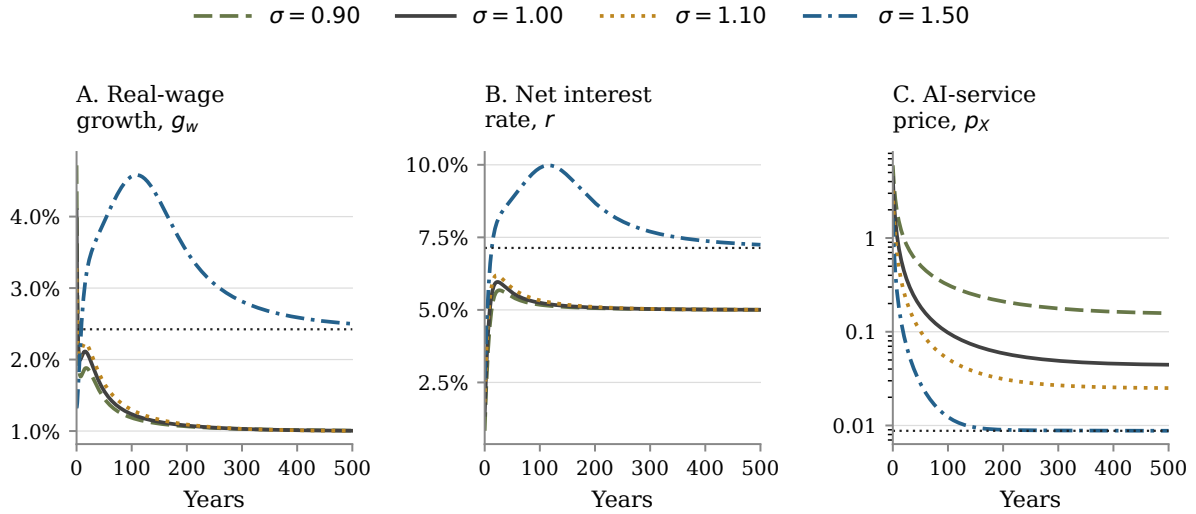


Figure 8: Wages, returns, and the AI-service price after the switch from the no-AI steady-state stocks. Each panel compares four admitted numerical equilibrium trajectories over 500 model years. Real-wage growth and the net interest rate use linear percentage scales; the level of p_X uses a logarithmic scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

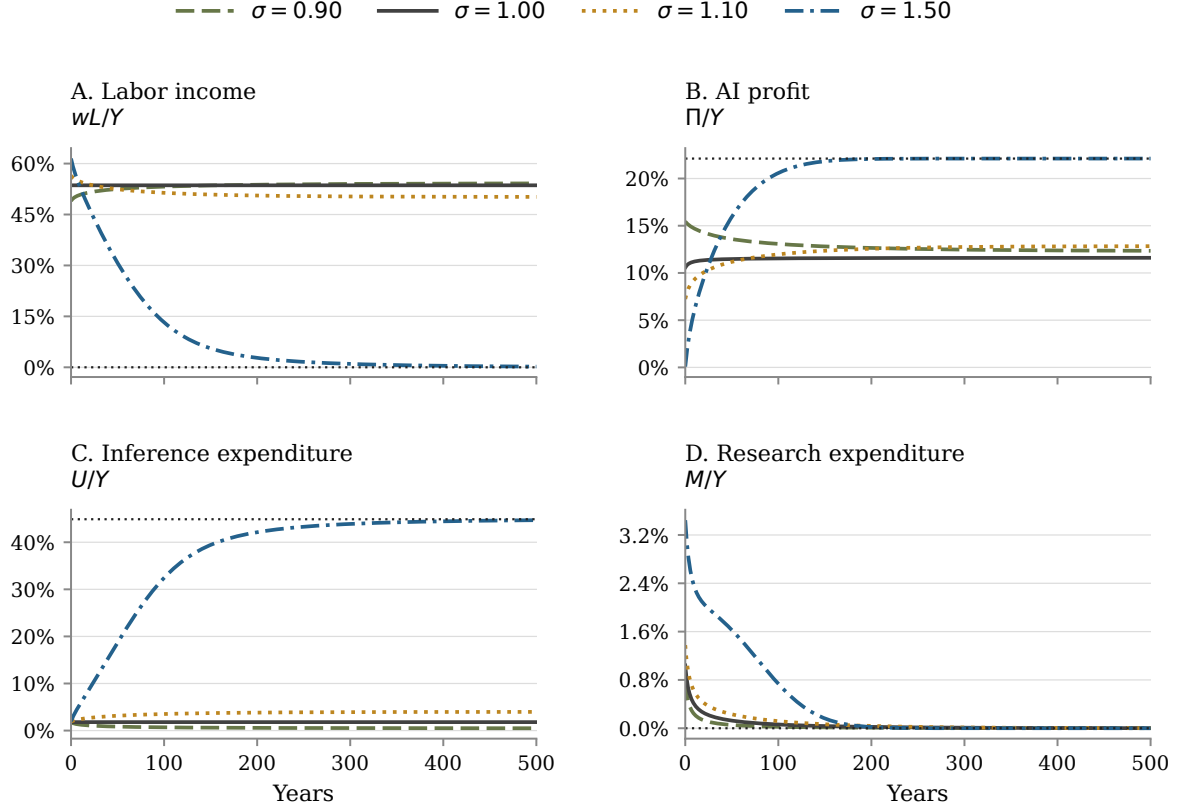


Figure 9: Distribution of the noncapital share of output after the switch from the no-AI steady-state stocks. Labor income, AI profit, inference expenditure, and research expenditure are divided by final output and sum to $1 - \alpha$. The omitted gross capital-income share is the constant α . Each panel compares four admitted numerical equilibrium trajectories over 500 model years. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

Low initial AI efficiency produces a sharp reallocation. At date zero, $X/(AL)$ grows between 30.5% and 79.0% across the four equilibria, while $K/(AL)$ falls between 5.7% and 8.1%. The net interest rate is initially between 0.83% and 2.59%, below its 5% no-AI value. These movements do not imply that access to an additional technology must lower current output. The experiment changes the CES weights from $(\omega_L, \omega_X) = (1, 0)$ to $(0.80, 0.20)$, so the old production technology does not remain available as an outside option. It should therefore be read as a change in technological regime from Ramsey initial stocks, not as a welfare comparison that holds the pre-change production set fixed.

The high-substitution economy subsequently separates from the other three. For $\sigma = 1.50$, labor initially receives 61.5% of output. Its share completes half of the decline toward zero after approximately 50 model years and falls to 0.2% by year 500. Output-per-person growth peaks at 6.38% around year 107, real-wage growth peaks at 4.58% around year 109, and the net interest rate peaks at 9.98% around year 116. By year 500, the three rates are 3.25%, 2.50%, and 7.25%, respectively, and continue toward their analytical limits. In the other three economies, output-per-

person and real-wage growth have returned to approximately 1% and the net interest rate to 5% by year 500. The same Ramsey initial capital stock can therefore lead either back to a labor-bottleneck regime or toward the AI-dominated regime; the difference is the elasticity and the upper bound relative to its elasticity-specific threshold.

6.5 Near-terminal paths

The initially high and declining growth rates of the three labor-bottleneck paths in the main comparison are not their long-run prediction. They partly reflect the distance between the common initial stocks in (60) and each economy's terminal allocation. To isolate terminal behavior while preserving the comparison across regimes, I solve an additional exercise for all four values of σ . I set

$$\frac{B_0}{\bar{B}} = 0.9999, \quad \frac{K_0}{A_0 N_0} = \frac{K}{AN} \Big|_{\text{terminal}} = \begin{cases} 6.5571, & \sigma = 0.90, \\ 9.0990, & \sigma = 1, \\ 12.2652, & \sigma = 1.10. \end{cases} \quad (66)$$

for the three labor-bottleneck paths. In the AI-dominated regime, $K/(AN)$ has no finite terminal value. Instead, $(\bar{B} - B)K$ converges to a positive constant $\bar{d}$, as derived in the proof of Proposition 1(iv). I initialize this product at its limiting value:

$$(\bar{B} - B_0)K_0 = \bar{d}, \quad K_0 = 4,101,704.23 \quad \text{when } \sigma = 1.50. \quad (67)$$

Consumption and the developer's shadow value are again selected by the BVP for each path. I use $B_0/\bar{B} = 0.9999$, rather than the upper bound itself, to begin sufficiently close to the analytical limits while preserving the interior research problem. For each labor-bottleneck path, setting capital at its terminal normalized value removes a separate capital-adjustment gap. For $\sigma = 1.50$, the corresponding stationary object is $(\bar{B} - B)K$, not $K/(AN)$, which explains the different capital rule in (67). These are diagnostic initial conditions chosen to display terminal behavior; they are not empirical calibrations. Because $B_0 < \bar{B}$ and research remains positive, these are not exact steady states at a finite date. They are equilibrium paths that begin near their respective terminal regimes. The initial capital condition varies across regimes because their convergent state variables differ. This exercise therefore isolates terminal behavior; it does not hold all initial stocks fixed.

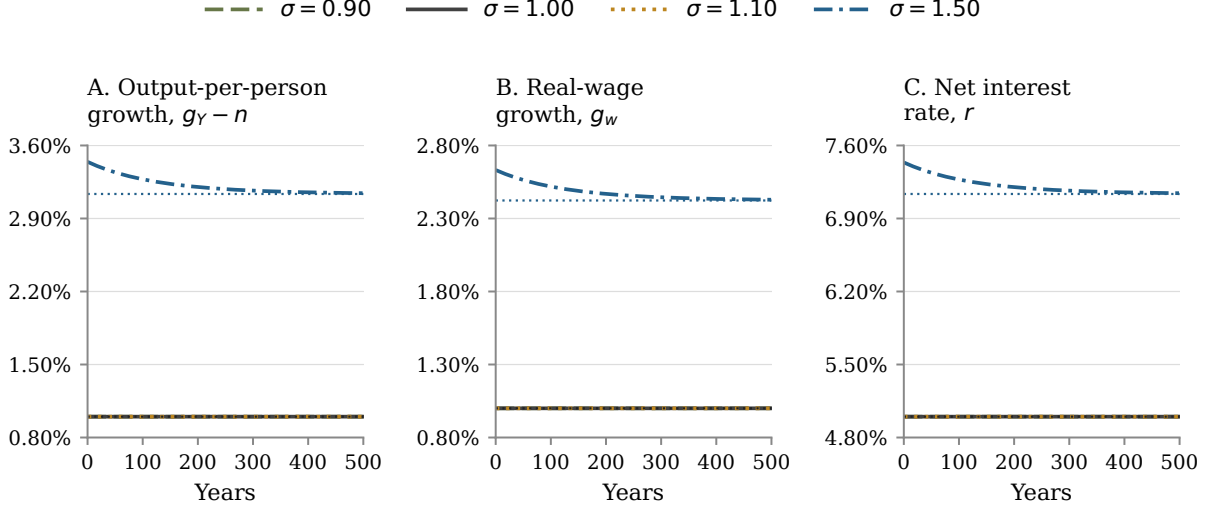


Figure 10: Near-terminal equilibrium paths. The panels report output-per-person growth, real-wage growth, and the net interest rate for four independently admitted numerical equilibria over 500 model years. Dark horizontal dotted lines mark the common labor-bottleneck limits; blue dotted lines mark the analytical limits for $\sigma = 1.50$.

Figure 10 removes the large date-zero movements of the three labor-bottleneck paths in the main comparison. Their output-per-person growth, real-wage growth, and net interest rate all begin within 0.001 percentage points of their analytical limits of 1%, 1%, and 5%. For $\sigma = 1.50$, the corresponding date-zero values are 3.443%, 2.632%, and 7.437%; by year 500 they have reached 3.143%, 2.429%, and 7.143%, close to their analytical limits of 3.135%, 2.423%, and 7.135%. The exercise confirms that the high initial growth rates in the main labor-bottleneck paths are transitional responses to the common starting stocks. It also displays the distinct AI-dominated long run without the much larger initial adjustment present in the main comparison.

6.6 A slower transition

The 50-year midpoint in the main comparison is a timing calibration rather than a prediction. To show how strongly the transition can stretch out, I recover the earlier calibration with

$$\chi = 0.01, \quad K_0 = 2.027733654, \quad B_0 = 0.443670932. \quad (68)$$

All other parameters, the common upper bound, and the four values of σ are unchanged. These initial stocks are the values implied by the earlier uncapped unit-elastic BGP. Together with the low value $\chi = 0.01$, they place the capped economy far from its upper bound and are chosen to ask whether the same long-run regimes can remain hidden for much longer. They are a sensitivity design, not estimates of the current world economy. Neither the initial consumption nor the initial shadow value is imported from the uncapped BGP. Both jump variables are solved again for each capped economy. The exercise changes both research productivity and the initial distance from the

upper bound, so it should not be interpreted as a one-parameter estimate of the effect of χ . Each of the four paths is independently re-solved and passes the same equilibrium-admission tests as the main comparison.

Figures 11–13 retain the variables, normalizations, scenario order, and scale conventions of the main figures. The only presentational change is the 4,000-year window needed to display the slower adjustment.

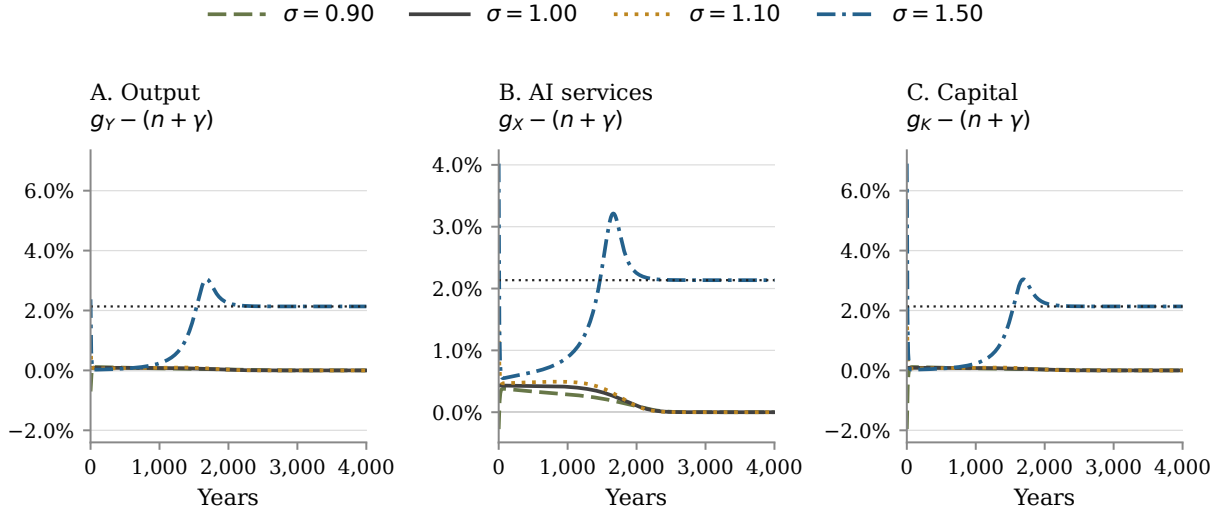


Figure 11: Growth of output, effective AI production services, and capital per unit of effective labor under the slower-transition calibration. Each panel compares the same four admitted numerical equilibrium trajectories over 4,000 model years. Panels A and C use the same linear percentage scale; Panel B uses a separate linear percentage range. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

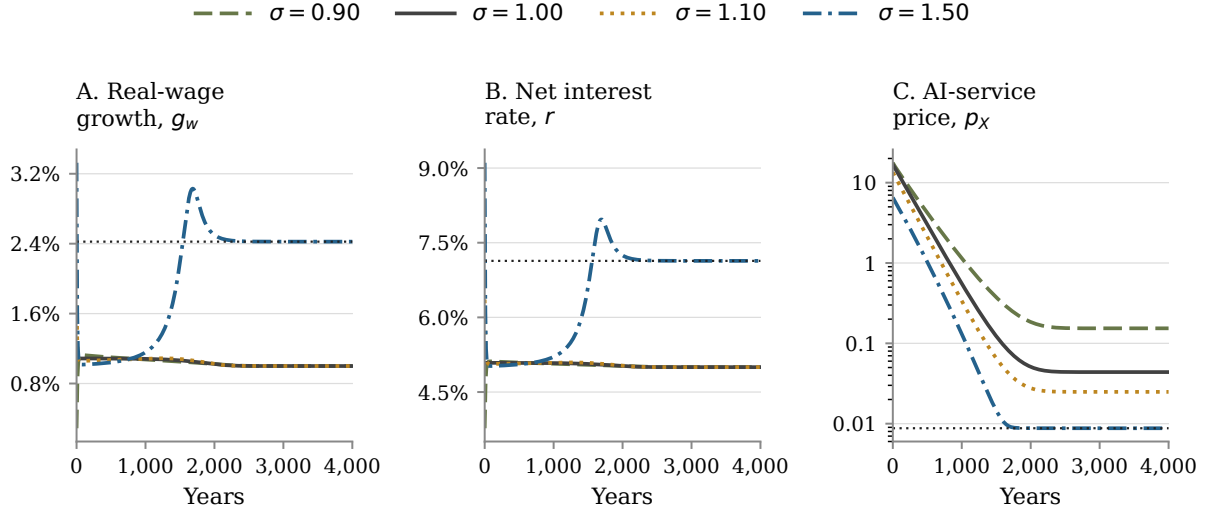


Figure 12: Wages, returns, and the AI-service price under the slower-transition calibration. Each panel compares the same four admitted numerical equilibrium trajectories over 4,000 model years. Real-wage growth and the net interest rate use linear percentage scales; the level of p_X uses a logarithmic scale. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.



Figure 13: Distribution of the noncapital share of output under the slower-transition calibration. Labor income, AI profit, inference expenditure, and research expenditure are divided by final output and sum to $1 - \alpha$. The omitted gross capital-income share is the constant α . Each panel compares the same four admitted numerical equilibrium trajectories over 4,000 model years. Thin horizontal dotted lines mark the analytical limits for $\sigma = 1.50$.

In the slow $\sigma = 1.50$ equilibrium, labor's income share completes only 10 percent of its decline after 657 model years, half after 1,295 years, and 90 percent after 1,605 years. At year 500, output per person grows at 1.062%, the real wage at 1.043%, and labor still receives 60.5% of output. By year 1,500, output-per-person growth has risen to 2.797%, while labor's share has fallen to 14.5%. The analytical limits are the same as in the main comparison because σ and $\bar{B}$ are unchanged; what changes radically is when the economy approaches them. The model can therefore generate a long interval in which little appears to happen before growth and the distribution of output change sharply. It does not identify which of the two timing calibrations is empirically more plausible.

6.7 Initial financing sensitivity

Moving the common initial stocks closer to the $\sigma = 1$ terminal regime materially reduces the initial financing requirement. In the reported $\sigma = 1.50$ path, research absorbs 12.0% of AI revenue, inference absorbs 41.4%, and the developer distributes the remaining 46.6% as profit. These quantities equal 1.92%, 6.60%, and 7.44% of final output, respectively.

This benign date-zero composition is not invariant to the initial stocks. In an admitted sensitivity path, I deliberately reuse the more distant stocks $K_0 = 2.0277$ and $B_0/\bar{B} = 0.26\%$ from the earlier numerical design. This choice isolates the financing consequences of beginning farther from the terminal regime. I then recalibrate χ to preserve the same 50-year transition midpoint as in the main $\sigma = 1.50$ path, so the comparison does not merely contrast different transition dates. The sensitivity path is a separately solved equilibrium, and its C_0 and q_0 are again selected by the BVP. Research expenditure initially equals 106.0% of AI revenue and profit is -40.5% of AI revenue. The household owner then finances the loss through an equity injection because current research raises B and the present value of future monopoly rents. The benchmark permits this intertemporal financing: it imposes neither a borrowing constraint nor a nonnegative-profit constraint on the developer. Adding such a constraint could delay or restrict the transition. The contrast therefore identifies initial technological distance and financing capacity as substantive determinants of the transition, not as innocuous plotting choices.

In the AI-dominated limit, inference absorbs 67% of AI revenue, research’s revenue share vanishes, and the developer retains 33%. Its net distribution therefore approaches 22.11% of final output. By contrast, at $\sigma = 1$ AI revenue is always 13.4% of output and inference absorbs 13.4% of that revenue. A vanishing research share does not mean that research stops: the constructed equilibria with a finite upper bound have a positive limiting level of research compute, while industry revenue continues to grow.

7 Conclusion

The endogenous growth model showed that the long-run economic role of human labor depends critically on whether AI can replace labor’s contribution to production. It characterizes how this substitution margin shapes economic growth, wages, interest rates, and the functional distribution of income when AI research is autonomous.

In the equilibria studied here with a finite upper bound, complementarity leaves human labor as a bottleneck. Improvements in AI efficiency raise productive capacity but do not permanently increase the growth of output per person or real wages beyond the exogenous rate of labor-augmenting technological progress. Labor therefore retains a positive share of income. The same broad conclusion holds under unit elasticity: AI can support sustained growth, but human labor remains economically essential and continues to receive a positive share of output.

The outcome changes when AI and labor are sufficiently substitutable. Gross substitution is necessary but not sufficient: labor remains a bottleneck while AI services are not productive enough to replace its contribution. Once AI becomes sufficiently efficient, however, capital and effective AI production services can expand faster than effective labor. Output per person and real wages then grow faster than in the labor-bottleneck regime, and the interest rate is higher.

Across the long-run regimes I characterize with a finite upper bound on AI efficiency, real wages grow faster than the exogenous rate only when labor’s income share vanishes. The declining share

does not generate faster wage growth. In the AI-dominated regime, output per person grows rapidly enough to raise real wages despite the shrinking fraction paid to labor. Because output per person grows even faster than the real wage, labor’s income share converges to zero. The gross capital share is fixed by the production technology, so this transformation reallocates the remainder of output from labor toward the AI industry. AI-industry revenue should not be confused with monopoly profit: it finances inference and research before the residual is distributed to the developer’s owner.

The uncapped unit-elastic benchmark shows why this qualification matters: continued AI-efficiency growth can raise output per person and real wages together faster than the exogenous rate while labor’s income share remains positive. With complementarity, by contrast, production per unit of effective labor remains bounded even without an upper bound on AI efficiency. This feasibility bound does not establish an equilibrium, but it rules out permanently faster growth of output per person than the exogenous rate.

The transition can be as important as the long-run outcome. In the calibrated AI-dominated path, labor completes half of its income-share decline after 50 model years while output and wages grow rapidly. This timing is illustrative: research productivity is not estimated, and changing it produces faster or slower equilibrium adjustments rather than merely relabeling time. The initial technological distance matters as well. The simulations nevertheless reveal signals before most of the decline in labor’s income share occurs: effective AI production services can grow rapidly, the interest rate can rise above the labor-bottleneck benchmark, and a positive output–wage growth gap shows that labor’s income share is already falling. Starting farther from the terminal regime can require the developer’s owner to finance large temporary losses against future monopoly rents.

The analysis points to three lines of future work. A task-based extension with heterogeneous workers, new human tasks, and a continuing role for human researchers could combine the displacement and reinstatement mechanisms of [Acemoglu and Restrepo \(2019\)](#), the labor-market transitions studied by [Korinek et al. \(2026\)](#), and the research-automation network of [Davidson et al. \(2026\)](#). An explicit upstream compute sector, alternative market structures, and heterogeneous asset ownership could build on [Korinek and McKelvey \(2026\)](#) and [Korinek and Vipra \(2025\)](#) and translate changes in functional income shares into household inequality and economic power. Finally, a more complete existence analysis could extend the uncapped result beyond unit elasticity, or make the upper bound on AI efficiency endogenous, and determine whether the resulting economy admits an equilibrium. This question is related to, but not resolved by, the explosive-growth mechanisms in [Nordhaus \(2021\)](#) and [Davidson et al. \(2026\)](#).

The central question is whether human labor remains sufficiently essential for wages to keep pace with the economy as a whole.

A Proofs

Lemma 1 (Static AI-service choice). *For $K, A, L, B > 0$, $0 < \alpha < 1$, $\sigma > 0$, and strictly positive CES weights that sum to one, problem (18) has a unique, strictly positive global maximizer. It is characterized by condition (19) and depends smoothly on the positive variables K, A, L, B , for each fixed σ .*

Proof of Lemma 1. Fix $K, A, L, B > 0$, let $R(X) = p_X(X)X$, and write

$$u = \frac{1}{\sigma}, \quad s = s_X, \quad e = u(1 - s) + \alpha s.$$

The demand elasticity in (15) and the CES-share derivative imply

$$R'(X) = p_X(1 - e), \quad \frac{ds}{d \ln X} = (1 - u)s(1 - s).$$

Let $m(X) = R'(X)$. Wherever marginal revenue is positive, direct differentiation gives

$$-\frac{Xm'(X)}{p_X} = e(1 - e) + (\alpha - u)(1 - u)s(1 - s) \equiv D(s). \quad (69)$$

If $(\alpha - u)(1 - u) \geq 0$, then $D(s) > 0$ immediately. If this product is negative, the parameter restrictions imply $0 < \alpha < u < 1$, and

$$D(s) = u(1 - u)(1 - s)^2 + [u^2 + 2\alpha - 3\alpha u]s(1 - s) + \alpha(1 - \alpha)s^2.$$

The middle coefficient is positive. For $u \leq 2/3$, this follows from $u^2 + \alpha(2 - 3u) > 0$. For $u > 2/3$, use $\alpha < u$ and $2 - 3u < 0$ to obtain

$$u^2 + \alpha(2 - 3u) > u^2 + u(2 - 3u) = 2u(1 - u) > 0.$$

Thus marginal revenue decreases strictly whenever it is positive. At a solution of $m(X) = 1/B$, its derivative is strictly negative. The implicit-function theorem therefore makes the maximizing service choice smooth in K, A, L, B wherever it exists.

For $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$. As $X \downarrow 0$, revenue is proportional to X^{1-u} when $\sigma > 1$, to X^β when $\sigma = 1$, and to $X^{1-\alpha}$ when $\sigma < 1$. Every exponent lies between zero and one, so $m(X) \rightarrow \infty$. As $X \rightarrow \infty$, marginal revenue converges to zero from above if $\sigma \geq 1$; if $\sigma < 1$, it eventually becomes negative. Hence $m(X) = 1/B$ has exactly one solution. Operating profit increases before this solution and decreases after it. The linear cost X/B dominates revenue as $X \rightarrow \infty$, so this solution is the unique global maximum. Equation (69) also proves the strict second-order condition:

$$e_X(1 - e_X) + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_X(1 - s_X) > 0. \quad (70)$$

□

Interior research. Lemma 1 establishes $X^* > 0$, and the envelope condition (20) gives $\pi_B = X^*/B^2 > 0$. A marginal increase in AI efficiency therefore raises current optimized profit and weakly raises future AI efficiency under a fixed research policy. At a differentiable optimum its current-value shadow price is strictly positive: $q > 0$. The part of the Hamiltonian that depends on M is

$$-M + q\chi B^\eta M^\eta \psi(B).$$

Its right derivative diverges as $M \downarrow 0$ because $q, B, \psi(B) > 0$ and $0 < \eta < 1$. Therefore $M = 0$ cannot maximize the Hamiltonian at a finite date.

Global developer verification. For a finite upper bound, define research depth as the monotone transformation

$$R_B(B) = -\bar{B} \log \left(1 - \frac{B}{\bar{B}} \right), \quad B(R_B) = \bar{B} \left(1 - e^{-R_B/\bar{B}} \right). \quad (71)$$

It maps $B \in [B_0, \bar{B})$ into $R_B \in [R_{B,0}, \infty)$, where $R_{B,t} = R_B(B_t)$ and $R_{B,0} = R_B(B_0)$, and rewrites the AI-efficiency law as

$$\dot{R}_B = \chi[B(R_B)M]^\eta. \quad (72)$$

Write $\pi_t(B)$ for optimized operating profit evaluated at the date- t values of K, A , and L , and define the shadow value of research depth as $q_{R,t} = q_t \psi(B_t)$. Holding this shadow value fixed, let

$$\widehat{\mathcal{H}}_t(\widetilde{R}_B) \equiv \max_{\widetilde{M} \geq 0} \left\{ \pi_t(B(\widetilde{R}_B)) - \widetilde{M} + q_{R,t} \chi[B(\widetilde{R}_B)\widetilde{M}]^\eta \right\}.$$

Lemma 2 (Global verification of the developer's research path). *Let $0 < \eta < 1$ and $\chi > 0$. Fix $0 < B_0 < \bar{B} < \infty$ and positive continuous paths for K, A , and L , and take the locally integrable interest-rate path r as given. Consider a feasible candidate $\{B_t, M_t\}_{t \geq 0}$ for problem (21), with continuously differentiable B_t , interior research $M_t > 0$, and a finite objective. Suppose there is a continuously differentiable costate q_t satisfying the research condition (22), the costate equation (23), and the transversality condition (25). If, for every $t \geq 0$ and every $\widetilde{R}_B \geq R_{B,0}$,*

$$\widehat{\mathcal{H}}_t(\widetilde{R}_B) \leq \widehat{\mathcal{H}}_t(R_{B,t}) + \widehat{\mathcal{H}}'_t(R_{B,t})(\widetilde{R}_B - R_{B,t}), \quad (73)$$

then the candidate is a global maximizer of the reduced developer problem. Admissible alternatives have $M_t \geq 0$, locally finite research expenditure, and the same initial state; the comparison also bounds the upper limit of their truncated discounted payoffs. Together with the unique static service choice characterized in Lemma 1, it is globally optimal for the developer.

Proof of Lemma 2. Fix a date and the candidate shadow price $q_{R,t} = q_t \psi(B_t) > 0$. Maximizing

the research-depth Hamiltonian over research expenditure gives

$$\hat{\mathcal{H}}_t(\tilde{R}_B) = \pi_t(B(\tilde{R}_B)) + \frac{1-\eta}{\eta} M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)}. \quad (74)$$

Here M_t is the candidate's research expenditure. The formula follows by maximizing $-M + q_{R,t}\chi[B(\tilde{R}_B)M]^\eta$ over $M \geq 0$ and using the candidate's research first-order condition. Its derivative at the candidate state is

$$\hat{\mathcal{H}}'_t(R_{B,t}) = \frac{\psi(B_t)}{B_t} (U_t + M_t). \quad (75)$$

Write $D_t = \exp(-\int_0^t r_s ds)$ and $\mu_t = D_t q_{R,t}$. The transformed costate equation is $\dot{\mu}_t = -D_t \hat{\mathcal{H}}'_t(R_{B,t})$. For any admissible alternative research policy and its state $\tilde{R}_{B,t}$, maximizing over its control first bounds its Hamiltonian by (74). Apply (73), subtract the candidate payoff, and integrate by parts. Equal initial states imply

$$\tilde{J}_T - J_T \leq -\mu_T(\tilde{R}_{B,T} - R_{B,T}) \leq \mu_T R_{B,T}.$$

Because

$$\psi(B_t)R_{B,t} = -(\bar{B} - B_t) \log \left(1 - \frac{B_t}{\bar{B}} \right) \leq B_t,$$

the last expression converges to zero by the developer's transversality condition. Taking the upper limit establishes global optimality. The argument requires a global upper tangent at the candidate state, not concavity at every counterfactual state. $\square$

A concavity test for Lemma 2. Fix a date t and an alternative AI-efficiency level $b \in [B_0, \bar{B}]$. Let $X_t^*(b)$ be the unique maximizer of problem (18), evaluated at K_t, A_t , and L_t , and let $s_{X,t}(b)$ and $e_{X,t}(b)$ be the corresponding values. Implicitly differentiating the monopoly condition (19), using Equations (10) and (15), gives

$$\varepsilon_{B,t}(b) \equiv \frac{\partial \ln X_t^*(b)}{\partial \ln b} = \frac{1 - e_{X,t}(b)}{e_{X,t}(b)[1 - e_{X,t}(b)] + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_{X,t}(b)[1 - s_{X,t}(b)]}. \quad (76)$$

This endogenous elasticity is not an additional model parameter. In what follows, suppress the date and counterfactual-state arguments and write

$$X^*(B) \equiv X_t^*(b), \quad \varepsilon_B \equiv \varepsilon_{B,t}(b).$$

Because $\pi_B = X^*(B)/B^2$,

$$\pi_{BB} = \frac{X^*}{B^3} (\varepsilon_B - 2). \quad (77)$$

A direct differentiation gives

$$\frac{\partial^2 \pi(B(R_B))}{\partial R_B^2} = \frac{X^*(B)\psi(B)}{B^3} \left[\psi(B)(\varepsilon_B - 2) - \frac{B}{\bar{B}} \right]. \quad (78)$$

The transformation (71) satisfies

$$B'(R_B) = \psi(B) > 0, \quad B''(R_B) = -\frac{\psi(B)}{\bar{B}} < 0.$$

Using $\pi_B = X^*(B)/B^2$ and (77) gives (78). Hence the algebraic condition

$$\psi(b)[\varepsilon_{B,t}(b) - 2] \leq \frac{b}{\bar{B}} \quad (79)$$

for every $t \geq 0$ and $b \in [B_0, \bar{B})$ makes optimized operating profit concave in research depth throughout the reachable state domain.

The research term in (74) is a positive multiple of $B(R_B)^\mu$, where $\mu = \eta/(1 - \eta) > 0$. Its curvature is

$$\frac{d^2 B(R_B)^\mu}{dR_B^2} = \mu B^{\mu-2} \psi(B) \left[(\mu - 1)\psi(B) - \frac{B}{\bar{B}} \right]. \quad (80)$$

Thus (79), together with

$$\frac{B_0}{\bar{B}} \geq \max \left\{ 0, \frac{2\eta - 1}{\eta} \right\}, \quad (81)$$

makes the maximized Hamiltonian concave throughout the reachable state domain and proves (73). If $\eta \leq 1/2$, condition (81) holds automatically. But for every fixed $0 < \eta < 1$ its right-hand side is strictly below one, so it also holds without this restriction when initial efficiency is sufficiently close to the upper bound. These curvature tests are sufficient, not necessary: they can fail even when the Hamiltonian has the required supporting tangent at the candidate state.

At unit elasticity, the concavity test has a simple primitive form. Define $\beta = (1 - \alpha)\omega_X$. The static monopoly solution implies $\pi_t(B) = a_t B^{\beta/(1-\beta)}$ for some $a_t > 0$. Because $B(R_B)$ is increasing and concave, $R_B \mapsto \pi_t(B(R_B))$ is globally concave whenever $\beta \leq 1/2$. In the benchmark calibration in Table 2, $\beta = 0.134$, so this sufficient condition holds with substantial slack. It is not imposed as an additional maintained assumption; outside this range, the support condition can instead be checked directly.

The AI-dominated continuation supplies an analytical long-run check on the support inequality. For $1 < \sigma \leq 1/\alpha$, the service elasticity defined in Equation (76) satisfies $\varepsilon_{B,t}(b) \leq 1/\alpha$ for every t and b . Suppressing these arguments, subtracting $\alpha(1 - e_X)$ from the denominator in that equation gives

$$(\sigma^{-1} - \alpha)(1 - s_X) [(1 - \sigma^{-1})(1 - s_X) + (\sigma^{-1} - \alpha)s_X] \geq 0.$$

Choose $B_* \in (B_0, \bar{B})$ with $B_*/\bar{B} > \max\{0, (1 - 2\alpha)/(1 - \alpha)\}$. Operating profit is then concave in research depth for $B \geq B_*$. If $\eta \leq 1/2$, the research term in (74) is concave there as well. For a

candidate with $B_t \geq B_*$, it consequently suffices to check

$$\widehat{\mathcal{H}}_t(R_{B,t}) - \widehat{\mathcal{H}}'_t(R_{B,t})(R_{B,t} - R_{B,0}) \geq \widehat{\mathcal{H}}_t(R_B(B_*)). \quad (82)$$

Above B_* the tangent bound follows from concavity. Below B_* the Hamiltonian is increasing in B , so its value is at most the right-hand side, whereas the tangent is minimized at $R_{B,0}$.

Along an AI-dominated continuation, the difference between the two sides of (82), divided by Y_t , converges to

$$\alpha(1 - \alpha) \left[1 - \left(\frac{B_*}{\bar{B}} \right)^{(1-\alpha)/\alpha} \right] > 0.$$

To obtain this limit, use $M/Y \rightarrow 0$, $\psi(B)R_B \rightarrow 0$, and $\pi(B_*)/Y \rightarrow \alpha(1 - \alpha)(B_*/\bar{B})^{(1-\alpha)/\alpha}$. Thus the global support condition holds sufficiently late on this continuation even when the stronger curvature condition fails at low counterfactual AI-efficiency levels. The finite transition still requires its own support check.

Proof of Proposition 1. The proof has three tasks. First, Lemma 1 gives a unique, smooth, globally optimal within-date service choice at given states. I combine this choice with the other equilibrium equations to identify candidate long-run values and verify their admissibility, including positive consumption. Finding these values does not yet establish the existence of a trajectory approaching them.

Second, I write the exact dynamic equations in coordinates with finite candidate limits. The linearization identifies the stable directions; I also check that they can accommodate every sufficiently nearby initial state. Counting stable directions alone would not establish this property. The stable-manifold and inverse-function theorems then select the initial consumption and shadow price for those states. They establish convergent solutions of the nonlinear equations for all future time, not merely solutions of their linear approximation.

Third, I reconstruct the original variables and verify feasibility, market clearing, and both transversality conditions. Household concavity and Lemma 2 establish optimality. For the developer, I verify the required curvature throughout the domain of alternative research policies reachable from the initial efficiency. The upper bound makes this possible for initial efficiency sufficiently close to it. These final checks turn the constructed solutions of the first-order conditions into equilibria. The result is local in initial states, not in the time horizon.

Step 1. Feasible efficiency and candidate long-run values.

Let $D = \bar{B} - B$. The state equation with a finite upper bound gives

$$\frac{\dot{D}}{D} = -\frac{\chi}{\bar{B}}(BM)^\eta.$$

Integration gives

$$\bar{B} - B_t = (\bar{B} - B_0) \exp \left\{ -\frac{\chi}{\bar{B}} \int_0^t (B_s M_s)^\eta ds \right\} > 0. \quad (83)$$

Local integrability makes the exponent finite at every finite date, so $D_t > 0$.

For the candidate limits, impose $B = \bar{B}$ only in the normalized boundary equations. Actual trajectories will satisfy $B < \bar{B}$ at every finite date.

Write $\theta = (1 - \alpha)/\alpha$. For $\sigma \neq 1$, let $\varphi = (\sigma - 1)/\sigma$ and define

$$m(s) = 1 - \frac{1-s}{\sigma} - \alpha s, \quad \mathcal{H}_\sigma(s) = \left[(1-\alpha)s m(s) \left(\frac{\omega_X}{s} \right)^{1/\varphi} \right]^\theta.$$

Here $m(s) = 1 - e_X$ is the marginal-revenue factor, not research compute. The static monopoly optimum has $m(s_X) > 0$ by Lemma 1. The CES share, monopoly condition, and production function, equations (28d), (28i), and (28c), imply

$$\frac{X}{AN} = \left[\frac{\omega_L s_X}{\omega_X (1 - s_X)} \right]^{1/\varphi}, \quad \frac{U}{Y} = (1 - \alpha) s_X m(s_X), \quad (84)$$

$$\frac{Y}{K} = B^\theta \mathcal{H}_\sigma(s_X). \quad (85)$$

For example, $Z = X(\omega_X/s_X)^{1/\varphi}$ and $X = B(1 - \alpha)s_X m(s_X)Y$ give the last identity directly.

If $0 < \sigma < 1$, positive marginal revenue requires

$$s > s_{\min} \equiv \frac{1 - \sigma}{1 - \alpha\sigma}.$$

On $(s_{\min}, 1)$, both $m(s)$ and $s^{-1/(\sigma-1)}$ increase strictly. Since

$$\mathcal{H}_\sigma(s)^{1/\theta} = (1 - \alpha)\omega_X^{1/\varphi} m(s) s^{-1/(\sigma-1)},$$

the share map increases from zero to $\mathcal{H}_\sigma(1) = [(1 - \alpha)^2 \omega_X^{1/\varphi}]^\theta$. Thus the equation

$$\frac{\rho + \gamma + \delta}{\alpha} = \bar{B}^\theta \mathcal{H}_\sigma(\bar{s}_X) \quad (86)$$

has exactly one interior solution if and only if $\bar{B} > \mathcal{B}(\sigma)$.

If $\sigma > 1$, the share map decreases strictly on $(0, 1)$. To verify the sign, put $a = \sigma^{-1} - \alpha$, so $m(s) = \varphi + as$. Its logarithmic derivative has the sign of $a(\sigma - 2)s - \varphi$. If $a(\sigma - 2) \leq 0$ this is negative. If $\sigma > 2$ and $a > 0$, $a(\sigma - 2) < (\sigma - 2)/\sigma < \varphi$; if $1 < \sigma < 2$ and $a < 0$, $a(\sigma - 2) < (1 - \sigma^{-1})(2 - \sigma) < \varphi$. The map therefore decreases from infinity to $\mathcal{H}_\sigma(1)$. Equation (86) has exactly one interior solution if and only if $\bar{B} < \mathcal{B}(\sigma)$. Since $\mathcal{R}(B) = \alpha B^\theta \mathcal{H}_\sigma(1) - \delta$, these calculations also derive (35)–(36). The interior-share condition is therefore $\mathcal{R}(\bar{B}) > \rho + \gamma$ for $\sigma < 1$, but $\mathcal{R}(\bar{B}) < \rho + \gamma$ for $\sigma > 1$.

For either interior-share case, let x, y, u, k, c denote X, Y, U, K, C divided by AN , respectively.

Construct the normalized terminal values sequentially:

$$\begin{aligned}
\bar{x} &= \left[\frac{\omega_L \bar{s}_X}{\omega_X (1 - \bar{s}_X)} \right]^{1/\varphi}, & \bar{u} &= \frac{\bar{x}}{\bar{B}}, \\
\bar{y} &= \frac{\bar{u}}{(1 - \alpha) \bar{s}_X m(\bar{s}_X)}, & \bar{k} &= \frac{\alpha \bar{y}}{\rho + \gamma + \delta}, \\
\bar{c} &= \bar{y} - \bar{u} - (\delta + n + \gamma) \bar{k}, & \bar{r} &= \rho + \gamma.
\end{aligned} \tag{87}$$

At $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$. The exact static solution is

$$y = \left[(\beta^2 B)^\beta k^\alpha \right]^{1/(1-\beta)}, \quad u = \beta^2 y, \quad x = Bu.$$

The equation $y/k = (\rho + \gamma + \delta)/\alpha$ has exactly one positive solution for every finite $\bar{B} > 0$:

$$\bar{k} = \left[(\beta^2 \bar{B})^\beta \left(\frac{\rho + \gamma + \delta}{\alpha} \right)^{\beta-1} \right]^{1/(1-\alpha-\beta)}. \tag{88}$$

Set $\bar{y} = [(\rho + \gamma + \delta)/\alpha] \bar{k}$, $\bar{u} = \beta^2 \bar{y}$, and $\bar{x} = \bar{B} \bar{u}$, with $\bar{c}, \bar{r}$ as in (87).

Consumption positivity does not require an additional restriction. Positive marginal revenue implies $0 < m(\bar{s}_X) < 1$. Since $0 < \bar{s}_X < 1$, in all three cases

$$\frac{\bar{u}}{\bar{y}} = (1 - \alpha) \bar{s}_X m(\bar{s}_X) < 1 - \alpha, \quad \frac{(\delta + n + \gamma) \bar{k}}{\bar{y}} = \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} < \alpha.$$

It follows that $\bar{c}/\bar{y} > 0$.

Step 2. The three positive-labor-share cases.

The cases in Proposition 1(i), 1(ii), and 1(iii) share one normalization. Divide capital, consumption, and the AI-efficiency shadow price by effective labor, and multiply the remaining efficiency gap by effective labor:

$$k = \frac{K}{AN}, \quad c = \frac{C}{AN}, \quad d = (\bar{B} - B)AN, \quad \tilde{q} = \frac{q}{AN}, \quad \tau = (AN)^{-1}. \tag{89}$$

Dividing by AN removes effective-labor growth. Multiplying the shrinking gap by AN allows it to converge to a positive value: $\dot{d}/d = n + \gamma - \dot{B}/(\bar{B} - B)$. The variable d measures the scaled gap, not its rate of closure. The constants $\bar{k}, \bar{d}$ are the limiting values of k, d ; they are determined by the model, not additional parameters. Also $\dot{\tau} = -(n + \gamma)\tau$, so $\tau \rightarrow 0$. These coordinates leave the model unchanged: $K = k/\tau$ and $B = \bar{B} - d\tau$ at every finite date.

Given k, B , solve the unique static monopoly problem for y, x, u , and set $r = \alpha y/k - \delta$. Define $a = \dot{B}/(\bar{B} - B)$, the proportional rate of closure of the AI-efficiency gap. The research condition gives

$$B = \bar{B} - d\tau, \quad M = \left[\tilde{q} d \chi \eta \frac{B^\eta}{\bar{B}} \right]^{1/(1-\eta)}, \quad a = \chi \frac{B^\eta M^\eta}{\bar{B}}.$$

Substituting (28k) and (29) gives the exact dynamic equations

$$\frac{\dot{k}}{k} = \frac{y - c - u - M\tau}{k} - \delta - n - \gamma, \quad (90a)$$

$$\frac{\dot{c}}{c} = r - \rho - \gamma, \quad (90b)$$

$$\frac{\dot{d}}{d} = n + \gamma - a, \quad (90c)$$

$$\frac{\dot{\tilde{q}}}{\tilde{q}} = r - \frac{x}{\tilde{q}B^2} - \eta \frac{ad\tau}{B} + a - n - \gamma, \quad (90d)$$

$$\dot{\tau} = -(n + \gamma)\tau. \quad (90e)$$

In particular, population growth cancels from the normalized consumption equation; it is not subtracted a second time.

Complete the terminal construction by setting

$$\bar{M} = \left[\frac{(n + \gamma)\bar{B}^{1-\eta}}{\chi} \right]^{1/\eta}, \quad \bar{\tilde{q}} = \frac{\bar{x}}{\bar{r}\bar{B}^2}, \quad \bar{d} = \frac{\bar{M}}{\eta(n + \gamma)\bar{\tilde{q}}}. \quad (91)$$

All are positive. Direct substitution shows that $(\bar{k}, \bar{c}, \bar{d}, \bar{\tilde{q}}, 0)$ is a fixed point of (90). This boundary point is a device for constructing paths with $\tau > 0$ and $B < \bar{B}$, not an admissible date-zero state itself.

I next verify saddle-path stability. Write $\mu = \eta/(1 - \eta)$, $r_k = \partial r / \partial \log k$, and $a_k = \partial[(y - u)/k] / \partial \log k + \bar{c}/\bar{k}$, evaluated at the fixed point. In $(\log k, \log c, \log d, \log \tilde{q}, \tau)$ coordinates the Jacobian has the two diagonal blocks

$$J_R = \begin{pmatrix} a_k & -\bar{c}/\bar{k} \\ r_k & 0 \end{pmatrix}, \quad J_B = \begin{pmatrix} -(n + \gamma)\mu & -(n + \gamma)\mu \\ (n + \gamma)\mu & \bar{r} + (n + \gamma)\mu \end{pmatrix}, \quad (92)$$

and the additional eigenvalue $-(n + \gamma)$. There can be forcing from τ to both blocks and from (k, c) to $(d, \tilde{q})$, but not in the reverse directions at the fixed point.

To check $r_k < 0$, let $s = s_X$, $u_\sigma = 1/\sigma$, $e = e_X$, and let $D(s)$ be the positive curvature expression in (69). Implicit differentiation of the monopoly condition gives

$$\frac{\partial \log X}{\partial \log K} = \frac{\alpha(1 - e)}{D(s)}, \quad r_k = (r + \delta)(1 - \alpha) \left[\frac{\alpha s(1 - e)}{D(s)} - 1 \right].$$

Moreover,

$$D(s) - \alpha s(1 - e) = (1 - s)\{u_\sigma(1 - e) + (\alpha - u_\sigma)(1 - u_\sigma)s\} > 0.$$

The sign is immediate when $u_\sigma \geq 1$ or $u_\sigma \leq \alpha$. For $\alpha < u_\sigma < 1$, the braces are linear in s , with positive endpoint values $u_\sigma(1 - u_\sigma)$ and $u_\sigma^2 + \alpha(1 - 2u_\sigma)$. The latter is positive if $u_\sigma \leq 1/2$; otherwise $\alpha < u_\sigma$ implies it exceeds $u_\sigma(1 - u_\sigma) > 0$. Thus $r_k < 0$ in every interior-share case.

Both blocks have negative determinants:

$$\det J_R = (\bar{c}/\bar{k})r_k < 0, \quad \det J_B = -(n + \gamma)\mu\bar{r} < 0.$$

There are therefore three stable and two unstable eigenvalues. The stable direction in J_R has a nonzero k component, and the stable direction in J_B has a nonzero d component. To verify that their number suffices to select the jump variables, consider a vector in the stable subspace whose state components (k, d, τ) are all zero. Its linearized τ path is zero. The homogeneous Ramsey block must then follow its stable direction; its zero initial k component forces its c component to be zero as well. With that forcing absent, the same argument in the research block forces $\tilde{q} = 0$. The projection of the three-dimensional stable subspace onto the three predetermined coordinates is therefore injective and hence invertible. This argument also covers repeated stable eigenvalues.

Lemma 1 makes the static solution smooth. The normalized vector field extends smoothly through $\tau = 0$, since $B = \bar{B} - d\tau$ remains positive nearby. The stable-manifold theorem and the inverse-function theorem therefore give a local graph $(c, \tilde{q})$ over (k, d, τ) ; see Dyatlov (2018). Choose a small neighborhood of the full fixed point and then a smaller initial-state neighborhood $\mathcal{U}_L$ of $(\bar{k}, \bar{d}, 0)$. Its constants depend on the parameters and the case under consideration. In original initial stocks the restriction is

$$(k_0, d_0, \tau_0) = \left(\frac{K_0}{A_0 N_0}, (\bar{B} - B_0)A_0 N_0, \frac{1}{A_0 N_0} \right) \in \mathcal{U}_L. \quad (93)$$

Closeness is required in all three coordinates; B_0 close to $\bar{B}$ alone is not sufficient. The point with $\tau = 0$ is a long-run limit, not a finite-date initial state. Every physical initial state satisfying (93) has a solution on this graph that stays in the full neighborhood for all future time and converges to the fixed point. It is unique among solutions with those properties. Positivity of $\tau, d, c, \tilde{q}, k$ and the original AI-efficiency law ensure $0 < B_t < \bar{B}$ and $M_t > 0$ at every finite date. Reconstruction gives the original initial stocks, the equilibrium equations, and the limiting rates in (37)–(39). Indeed, the normalized vector field tends to zero at its fixed point, so the time derivatives of its smooth static ratios tend to zero as well. This step, not convergence of ratios alone, establishes convergence of the growth rates. Optimality and transversality are verified below.

Step 3. The AI-dominated case.

Let $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$. Here $\varphi = (\sigma - 1)/\sigma \in (0, 1)$. Since capital outgrows effective labor, $K/(AN)$ has no finite limit. Instead normalize consumption and the shadow value by capital and scale the efficiency gap by capital:

$$h = (AN/K)^\varphi, \quad d = (\bar{B} - B)K, \quad c = C/K, \quad v = q/K, \quad \tau = (AN)^{-1}.$$

Within this construction, y, x, u denote $Y/K, X/K, U/K$, respectively. The constant $\bar{d}$ now denotes

the limit of $(\bar{B} - B)K$, not of $(\bar{B} - B)AN$; no original state or control is redefined. Then

$$B = \bar{B} - d\tau h^{1/\varphi}, \quad M = [vd\chi\eta B^\eta/\bar{B}]^{1/(1-\eta)}, \quad a = \chi B^\eta M^\eta/\bar{B}, \quad r = \alpha y - \delta.$$

Define $g_K = y - c - u - M\tau h^{1/\varphi} - \delta$. The exact equations are

$$\dot{h} = \varphi h(n + \gamma - g_K), \tag{94a}$$

$$\dot{d}/d = g_K - a, \tag{94b}$$

$$\dot{c}/c = n + r - \rho - g_K, \tag{94c}$$

$$\dot{v}/v = r - x/(vB^2) - \eta ad\tau h^{1/\varphi}/B + a - g_K, \tag{94d}$$

$$\dot{\tau} = -(n + \gamma)\tau. \tag{94e}$$

The CES term divided by capital is $[\omega_L h + \omega_X x^\varphi]^{1/\varphi}$. At $h = 0$ the monopoly equation has the strictly positive curvature $D(1) = \alpha(1 - \alpha)$. Its implicit static solution is therefore smooth in (h, B) near $(0, \bar{B})$. For the remaining terms, replacing $h^{1/\varphi}$ by $\max\{h, 0\}^{1/\varphi}$ outside the physical domain gives a C^1 extension because $1/\varphi > 1$. The C^1 stable-manifold theorem applies; see [Feldman \(2021, p. 7\)](#). The extension is only a mathematical device and does not change any equation on $h \geq 0$.

The limiting interest and aggregate growth rates are

$$\bar{r} = \mathcal{R}(\bar{B}), \quad \bar{g} = n + \bar{r} - \rho > n + \gamma. \tag{95}$$

At $h = \tau = 0$, the terminal constants are

$$\begin{aligned} \bar{y} &= \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}, & \bar{u} &= (1 - \alpha)^2 \bar{y}, & \bar{x} &= \bar{B} \bar{u}, \\ \bar{r} &= \mathcal{R}(\bar{B}), & \bar{g} &= n + \bar{r} - \rho, & \bar{c} &= \alpha(1 - \alpha)\bar{y} + \rho - n > 0, \\ \bar{v} &= \bar{x}/(\bar{r} \bar{B}^2), & \bar{M} &= [\bar{g} \bar{B}^{1-\eta}/\chi]^{1/\eta}, & \bar{d} &= \bar{M}/(\eta \bar{g} \bar{v}). \end{aligned} \tag{96}$$

The threshold implies $\bar{g} > n + \gamma > 0$ and $\bar{r} > \rho + \gamma > 0$, so all constants are positive. Substitution verifies the fixed point $(0, \bar{d}, \bar{c}, \bar{v}, 0)$.

The h and τ eigenvalues are $\varphi(n + \gamma - \bar{g}) < 0$ and $-(n + \gamma) < 0$. The first equals $\varphi[\rho + \gamma - \mathcal{R}(\bar{B})]$. It becomes zero at $\mathcal{R}(\bar{B}) = \rho + \gamma$, which is why this stable-manifold argument does not cover the threshold. The consumption-ratio eigenvalue is $\bar{c} > 0$. The remaining block in $(\log d, \log v)$ is

$$\begin{pmatrix} -\bar{g}\mu & -\bar{g}\mu \\ \bar{g}\mu & \bar{r} + \bar{g}\mu \end{pmatrix}, \quad \det = -\bar{g}\mu\bar{r} < 0.$$

The linearized system is triangular in the order $(h, \tau), c, (d, v)$. A stable vector with zero state components (h, d, τ) has zero h, τ paths; its consumption component must be zero because the unforced consumption equation is unstable. The research block then has zero d component and hence zero v component, as in the previous construction. Thus the three-dimensional stable subspace projects

invertibly onto (h, d, τ) . The same graph argument gives a neighborhood $\mathcal{U}_A$ of $(0, \bar{d}, 0)$ such that the initial-state condition is

$$\left[\left(\frac{A_0 N_0}{K_0} \right)^\varphi, (\bar{B} - B_0) K_0, \frac{1}{A_0 N_0} \right] \in \mathcal{U}_A. \quad (97)$$

It constructs a convergent trajectory for every admissible initial state in this neighborhood, with a unique nearby convergent choice of C_0, q_0 and positive quantities at every finite date.

The normalized vector field tends to zero, so the time derivatives of the smooth positive ratios Y/K , X/K , and C/K tend to zero. Consequently $g_Y, g_X, g_C, g_K \rightarrow \bar{g}$. Also $s_X \rightarrow 1$ and $1 - s_X$ is asymptotically proportional to $(AN/X)^\varphi$. The wage equation gives

$$g_w \rightarrow \bar{g} - n + \varphi(n + \gamma - \bar{g}) = \gamma + \frac{\bar{r} - \rho - \gamma}{\sigma}.$$

This proves the growth and distribution limits in Proposition 1(iv).

Step 4. Optimality and transversality.

In either construction let $\bar{g}$ denote the terminal aggregate growth rate: $n + \gamma$ in the positive labor-share case and $n + \bar{r} - \rho$ in the AI-dominated case. The constructions imply

$$a \rightarrow \bar{g}, \quad M \rightarrow \left[\frac{\bar{g} \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta} > 0, \quad g_q \rightarrow \bar{g}, \quad g_B \rightarrow 0, \quad M/Y \rightarrow 0.$$

Both transversality expressions in Definition 1 have limiting logarithmic growth

$$-\rho + n + g_K - g_C \rightarrow n - \rho < 0, \quad -r + g_q + g_B \rightarrow \bar{g} - \bar{r} = n - \rho < 0. \quad (98)$$

They therefore converge to zero, rather than merely remaining bounded.

Household utility is finite because $\log(C/N)$ grows at most linearly and $\rho > n$, as imposed in the household problem. Log utility is concave, the household budget is linear at given prices and distributions. More explicitly, let $\lambda_t = e^{-\rho t} N_t / C_t$, so $\dot{\lambda}_t = -r_t \lambda_t$ by (5). Concavity of log utility and equal initial capital give, for any admissible alternative $(\tilde{C}, \tilde{K})$,

$$\int_0^T e^{-\rho t} N_t \log \frac{\tilde{C}_t}{C_t} dt \leq \int_0^T \lambda_t (\tilde{C}_t - C_t) dt = -\lambda_T (\tilde{K}_T - K_T) \leq \lambda_T K_T \rightarrow 0.$$

Here $\tilde{K}_T \geq 0$ and the last limit is the household TVC. Final-good optimality follows from concavity and constant returns. The pointwise monopoly optimum is global by Lemma 1.

For the dynamic developer, feasibility and first-order conditions still require a sufficiency argument. Let $S_t = A_t N_t$ in the positive-share construction and $S_t = K_t$ in the AI-dominated construction, and let $D_t = \exp(-\int_0^t r_s ds)$. Homogeneity of the static problem and compactness of the normalized state neighborhood give a constant C_π such that $0 \leq \pi_t(b) \leq C_\pi S_t$ for every date and every $b \in [B_0, \bar{B}]$. Since $d \log(D_t S_t) / dt \rightarrow \bar{g} - \bar{r} = n - \rho < 0$, $\int_0^\infty D_t S_t dt < \infty$. Thus discounted

operating profit is uniformly bounded above across all alternative research policies. The candidate's objective is finite, because its research compute is bounded and $r_t \rightarrow \bar{r} > 0$. An alternative with infinite discounted research expenditure has payoff $-\infty$ and cannot improve on it.

It remains to verify the curvature conditions (79) and (81) for every date and every alternative AI-efficiency level in $[B_0, \bar{B})$, not just on the selected path. First fix a compact normalized-state neighborhood and an interval of counterfactual efficiencies $[b_*, \bar{B}]$, with $0 < b_* < \bar{B}$. In the positive labor-share construction, $K/(AN)$ stays in a compact neighborhood of $\bar{k} > 0$. Smoothness of the static monopoly solution bounds $\varepsilon_{B,t}(b)$ uniformly for these states and for AI-efficiency levels near $\bar{B}$. In the AI-dominated construction, h stays near zero, and the same uniform bound follows from the smooth static limit $\varepsilon_{B,t}(b) \rightarrow 1/\alpha$. Thus in either case there is a finite constant E with $\varepsilon_{B,t}(b) - 2 \leq E$ throughout that state domain. This bound is chosen before shrinking the initial-state neighborhood. Because $\mu = \eta/(1 - \eta)$ is finite for each $0 < \eta < 1$, choose $b_{**} \in [b_*, \bar{B})$ sufficiently close to $\bar{B}$ that

$$(1 - b_{**}/\bar{B}) \max\{E, \mu - 1, 0\} < b_{**}/\bar{B}.$$

Shrink the open neighborhood $\mathcal{U}_L$ or $\mathcal{U}_A$ so that $B_0 > b_{**}$. This is possible because $B_0 = \bar{B} - d_0\tau_0$ in the first construction and $B_0 = \bar{B} - d_0\tau_0 h_0^{1/\varphi}$ in the second. Every feasible research policy has $b \geq B_0$, so the two curvature conditions hold throughout the entire alternative-state domain at every date. Lemma 2 and the verified TVC therefore establish global developer optimality without imposing $\eta \leq 1/2$. No curvature property of the candidate is being assumed: it follows from the primitives and the chosen initial-state neighborhood.

All quantities remain finite at every finite date, and market clearing and the original initial conditions hold by reconstruction. The paths are therefore equilibria, not merely solutions of the canonical equations. The result is local only in its set of initial stocks. For $\sigma > 1$, at $\bar{B} = \mathcal{B}(\sigma)$ the interior-share terminal point reaches the boundary and a stable eigenvalue in the boundary normalization becomes zero; this proof does not apply. For $\sigma < 1$ and $\bar{B} \leq \mathcal{B}(\sigma)$, it does not rule out equilibria with different long-run ratios. Nor does it imply existence at arbitrary initial stocks or in the uncapped limit.

Step 5. Initial-state sets and the four conclusions.

The preimage of each open neighborhood under its continuous initial-state map, restricted to $A_0, N_0, K_0 > 0$ and $0 < B_0 < \bar{B}$, is open in the original initial stocks. It is nonempty. In the positive-labor-share cases, choose k_0, d_0 close to $\bar{k}, \bar{d}$ and $\tau_0 > 0$ sufficiently small. Then $A_0 N_0 = 1/\tau_0$, $K_0 = k_0/\tau_0$, and $B_0 = \bar{B} - d_0\tau_0$ are admissible. In the AI-dominated case, choose d_0 near $\bar{d}$ and $h_0, \tau_0 > 0$ sufficiently small. The inverse relations are $A_0 N_0 = 1/\tau_0$, $K_0 = 1/(\tau_0 h_0^{1/\varphi})$, and $B_0 = \bar{B} - d_0\tau_0 h_0^{1/\varphi}$. Any positive factorization of $A_0 N_0$ supplies A_0, N_0 . These choices also satisfy the strict efficiency restriction in Step 4.

For each initial state in the resulting set, the local graph selects $C_0, q_0 > 0$. These jump values are unique among the trajectories that remain near the indicated normalized limit and converge to

it. The argument does not assert global uniqueness.

For $\sigma < 1$ with $\bar{B} > \mathcal{B}(\sigma)$, and for $\sigma > 1$ with $\bar{B} < \mathcal{B}(\sigma)$, Step 1 gives a unique interior $\bar{s}_X$. Step 2 then yields $g_Y - n \rightarrow \gamma$, $g_w \rightarrow \gamma$, and $r \rightarrow \rho + \gamma$. At $\sigma = 1$, the exact CES limit gives $s_X = \omega_X$ at every date, and the same construction works for every finite positive upper bound. These establish the first three cases of the proposition. Step 3 gives $s_X \rightarrow 1$, $r \rightarrow \mathcal{R}(\bar{B})$, $g_Y - n \rightarrow \mathcal{R}(\bar{B}) - \rho$, and the wage-growth limit in (41), establishing the fourth case. Step 4 verifies the agent-optimality and transversality requirements of Definition 1 for all these trajectories. $\square$

Research expenditure and finite-horizon value. The first-order and transversality conditions do not by themselves establish that the developer's objective has a finite value or that a candidate policy is globally optimal. The result below addresses the value on a fixed finite horizon, taking the paths of K, A, L , and r as given. I retain $0 < \eta < 1$ from the AI-efficiency law and distinguish the uncapped cases $\eta < \alpha$, $\eta = \alpha$, and $\eta > \alpha$. Global optimality is addressed separately in Lemma 2.

Proposition 5 (Large-scale research expenditure). *Fix a finite horizon, an initial AI-efficiency level, positive continuous paths for K, A, L , and an integrable path for r . In the uncapped economy with $\bar{B} = \infty$, if $0 < \eta < \alpha < 1$, the developer's truncated value is finite and its objective tends to minus infinity as total research expenditure diverges. If $\sigma > 1$ and $\eta > \alpha$, fixed-duration expansions of research compute make the truncated value unbounded above. Consequently, there is no uncapped infinite-horizon equilibrium with a finite-valued developer optimum when $\sigma > 1$ and $\alpha < \eta < 1$. This nonexistence result does not require a finite-time singularity. At $\eta = \alpha$, the leading benefit and cost terms have the same order, so their coefficients determine the result. For a fixed finite upper bound, the truncated problem instead has a finite value and attains its maximum for every $0 < \eta < 1$.*

The uncapped result compares the cost of research with its effect on future operating profit. Expenditure first raises AI efficiency; higher AI efficiency then raises the profit from supplying effective AI production services. The proof bounds these two effects in turn and compares their combined growth with the linear cost of research.

Proof of Proposition 5. I first consider the uncapped economy. Define

$$\theta = \frac{1 - \alpha}{\alpha}, \quad S = \int_0^T M_t dt.$$

Here S is total research expenditure over the fixed horizon $[0, T]$. For an admissible research policy, let its discounted net payoff be

$$J_T(M) = \int_0^T \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt.$$

For every admissible policy, B is nondecreasing. The AI-efficiency law implies

$$\frac{d}{dt}B_t^{1-\eta} = (1-\eta)\chi M_t^\eta.$$

Integration and Hölder's inequality give

$$\begin{aligned} B_t^{1-\eta} &= B_0^{1-\eta} + (1-\eta)\chi \int_0^t M_s^\eta ds, \\ \int_0^T M_t^\eta dt &\leq T^{1-\eta} S^\eta. \end{aligned}$$

Because $0 < \eta < 1$, spreading a given expenditure evenly over $[0, T]$ maximizes the integral of M^η . Hence there is a constant $C_B > 0$, independent of the research policy, such that

$$\max_{0 \leq t \leq T} B_t \leq C_B \left(1 + S^{\eta/(1-\eta)}\right), \quad (99)$$

The same bound covers arbitrarily concentrated controls because, on a support of length $d \leq T$, $\int M_t^\eta dt \leq d^{1-\eta} S^\eta$.

Let $\bar{K} = \max_{t \leq T} K_t$ and $\bar{\mathcal{L}} = \max_{t \leq T} A_t L_t$. The weighted power-mean inequality gives $Z \leq AL + X$. The final-good firm's demand condition therefore implies

$$p_X X \leq (1-\alpha)\bar{K}^\alpha \left(\bar{\mathcal{L}}^{1-\alpha} + X^{1-\alpha}\right).$$

Write the right-hand side as $a + bX^{1-\alpha}$, with $a, b > 0$ determined by the given paths and production parameters. The first-order condition for maximizing this revenue bound net of X/B is $b(1-\alpha)X^{-\alpha} = 1/B$, and its maximized value is

$$\max_{X \geq 0} \left\{ a + bX^{1-\alpha} - \frac{X}{B} \right\} = a + \alpha b [b(1-\alpha)B]^\theta.$$

There is therefore a constant $C_\pi > 0$, independent of the research policy, such that

$$\pi(B) \leq C_\pi (1 + B^\theta). \quad (100)$$

Combining the AI-efficiency and profit bounds shows that operating profit grows no faster than a constant plus a term proportional to

$$S^{\eta(1-\alpha)/[\alpha(1-\eta)]}, \quad (101)$$

where the exponent is the product of $\eta/(1-\eta)$, from expenditure to AI efficiency, and $(1-\alpha)/\alpha$, from AI efficiency to profit. This is an upper bound on the order of growth, not an exact profit function.

On a finite horizon, integrability of r bounds the discount factor above and away from zero. Let $\underline{D} > 0$ be its minimum on $[0, T]$. Combining (99) and (100) yields, for constants $C_0, C_1 > 0$

independent of the research policy,

$$J_T(M) \leq C_0 + C_1 S^p - \underline{D}S, \quad p = \frac{\eta(1-\alpha)}{\alpha(1-\eta)}. \quad (102)$$

The exponent satisfies $p < 1$ exactly when $\eta < \alpha$. In this case, the benefit grows at most sublinearly in expenditure while its cost grows linearly. The bound tends to minus infinity as $S \rightarrow \infty$, so the value is finite and the objective is coercive.

An upper bound with $p > 1$ does not by itself prove unboundedness. For the converse, I construct research policies with unbounded payoffs. Suppose $\sigma > 1$. As $B \rightarrow \infty$, optimized operating profit satisfies

$$\pi(B) \sim \kappa_X K B^\theta$$

for a constant $\kappa_X > 0$, uniformly when K, A , and L remain in compact subsets of the positive orthant. Fix $\Delta \in (0, T)$, choose $M = \mathcal{M}$ on $[0, \Delta]$, and choose $M = 0$ afterward. Then

$$B_{\mathcal{M}}(t)^{1-\eta} = B_0^{1-\eta} + (1-\eta)\chi\mathcal{M}^\eta \min\{t, \Delta\}.$$

Choosing X optimally at every date gives

$$J_T(\mathcal{M}) = c_\pi \mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]} - c_M \mathcal{M} + o\left(\mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]}\right), \quad (103)$$

with $c_\pi, c_M > 0$. The exponent exceeds one when $\eta > \alpha$, so this family makes the truncated value unbounded above. At equality, the exponent is one and the sign of the leading coefficient determines this family's result.

To obtain the infinite-horizon nonexistence conclusion, suppose instead that an uncapped equilibrium has a developer policy $\widehat{M}$, an associated efficiency path $\widehat{B}$, and a finite discounted value $\widehat{V}_0$. Hold its paths of K, A, L , and r fixed, as required by the developer's problem (21). On $[0, T]$, use the burst policy above; after T , resume $\widehat{M}$. For a sufficiently large burst, $B_{\mathcal{M}}(T) > \widehat{B}(T)$. Under the uncapped law, the difference $B_{\mathcal{M}}(t)^{1-\eta} - \widehat{B}(t)^{1-\eta}$ is then constant and positive for $t \geq T$, since the two policies use the same research spending. Operating profit is increasing in B by equation (20), so subsequent net profit is no lower than under the candidate policy. Let $\widehat{J}_T$ be the candidate's discounted net profit on $[0, T]$ and let $J_\infty(\mathcal{M})$ denote the payoff from the deviating policy, including its continuation. Then

$$J_\infty(\mathcal{M}) \geq J_T(\mathcal{M}) + \widehat{V}_0 - \widehat{J}_T \longrightarrow +\infty.$$

The continuation payoff is well defined as a finite value or $+\infty$: its difference from the candidate's continuation payoff is the integral of a nonnegative operating-profit gain. Each deviation has finite research expenditure on every finite interval and hence finite B there by (99). Every finite-valued candidate can therefore be improved, contradicting developer optimality. The argument excludes a finite-valued optimum; it does not treat an infinite payoff as an attained firm value or assume that a trajectory reaches a singularity.

For a finite upper bound, the invariance result above confines B to the compact interval $[B_0, \overline{B}]$. Optimized operating profit is continuous in B and uniformly bounded on the fixed horizon. Hence $J_T(M) \leq C_0 - \underline{D}S$, so every maximizing sequence has bounded total expenditure. Set $v = M^\eta$ and $p = 1/\eta > 1$. The control cost is v^p , while the state equation is affine in v :

$$\dot{B} = \chi B^\eta \left(1 - \frac{B}{\overline{B}}\right) v.$$

A maximizing sequence is bounded in the reflexive space $L^p[0, T]$. A weakly convergent subsequence and the associated uniformly convergent state paths preserve feasibility. The operating-profit integral is continuous under uniform state convergence, while $\int v^p$ is weakly lower semicontinuous. The objective is therefore weakly upper semicontinuous and attains its maximum. $\square$

The condition $\eta < \alpha$ prevents arbitrarily large research expenditure from being optimal on a fixed horizon in the uncapped economy. It is a sufficient restriction, not a necessary one for every value of σ , and finite-horizon existence with a finite upper bound does not require $\eta < \alpha$. Neither the bound for $\eta < \alpha$ nor finite-horizon developer existence with a finite upper bound establishes an equilibrium: the paths of K, A, L , and r were taken as given. In contrast, for $\sigma > 1$ and $\eta > \alpha$, the profitable-deviation argument rules out an uncapped equilibrium with a finite-valued developer optimum at any such candidate paths. The quantitative analysis uses $\eta = 0.20 < \alpha = 0.33$, for which the expenditure exponent is approximately 0.508; the value of η is illustrative rather than estimated.

Proof of Proposition 2. Equations (43) and (44) follow because $\sigma/(\sigma-1) < 0$. Define $k = K/(AL)$. The resource constraint (29a), nonnegative consumption and compute expenditure, and $L = N$ imply

$$\dot{k} \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} k^\alpha - (\delta + n + \gamma)k. \quad (104)$$

Since $\delta + n + \gamma > 0$ and $0 < \alpha < 1$, the right-hand side is negative above a finite positive level. In particular,

$$k_t \leq \max \left\{ k_0, \left[\frac{\omega_L^{\sigma(1-\alpha)/(\sigma-1)}}{\delta + n + \gamma} \right]^{1/(1-\alpha)} \right\}.$$

Equation (44) then bounds $Y/(AL)$ uniformly. Since $A_t = A_0 e^{\gamma t}$ and $L = N$, Y/N is bounded above by a constant times $e^{\gamma t}$. Taking logarithms, dividing by t , and taking the upper limit gives (45).

The same bounds keep K and Y finite on every bounded time interval. For any such horizon T , integrating the resource constraint gives

$$\int_0^T M_t dt \leq K_0 + \int_0^T Y_t dt < \infty.$$

The uncapped research law and Hölder's inequality imply

$$\begin{aligned} B_T^{1-\eta} &= B_0^{1-\eta} + (1-\eta)\chi \int_0^T M_t^\eta dt \\ &\leq B_0^{1-\eta} + (1-\eta)\chi T^{1-\eta} \left(\int_0^T M_t dt \right)^\eta < \infty. \end{aligned} \quad (105)$$

Thus B cannot diverge at a finite date either. This argument bounds stocks and output; it is not a regularity theorem for all instantaneous control choices or an equilibrium-existence proof.

There is also no finite-horizon unbounded-payoff argument of the kind used for substitution in Proposition 5. At given positive continuous paths of K, A, L on $[0, T]$, optimized operating profit satisfies

$$0 \leq \pi_t(B) \leq p_X X \leq (1-\alpha)Y \leq (1-\alpha)\omega_L^{\sigma(1-\alpha)/(\sigma-1)} K_t^\alpha (A_t L_t)^{1-\alpha}.$$

This upper bound is independent of the developer's choice of B . For a given locally integrable r , the discount factor $D_t = \exp(-\int_0^t r_s ds)$ has a positive minimum on $[0, T]$. Discounted operating revenue is bounded, whereas the cost of research tends to infinity with $\int_0^T M_t dt$. The developer's truncated supremum is therefore finite, and net payoff tends to minus infinity as total research expenditure diverges. This conclusion holds for every $0 < \eta < 1$, but does not prove that an infinite-horizon optimum exists. $\square$

Proof of Proposition 3. All auxiliary variables below are confined to this proof. Write $x = X/(AL)$, $k = K/(AL)$, $c = C/(AL)$, $y = Y/(AL)$, $u = U/(AL)$, $m = M/(AL)$, $d = \rho - n > 0$, and $\nu = \delta + n + \gamma$. No convergence of a growth rate or allocation ratio is assumed.

Step 1. Static optimality gives a limiting production function. Put $\varphi = (\sigma - 1)/\sigma < 0$ and $f(x) = [\omega_L + \omega_X x^\varphi]^{(1-\alpha)/\varphi}$, so $y = k^\alpha f(x)$. The monopoly condition (19) requires $e_X = (1 - s_X)/\sigma + \alpha s_X < 1$. Thus

$$s_X > s_\infty := \frac{1-\sigma}{1-\alpha\sigma}, \quad 0 < x < x_\infty := \left[\frac{\omega_L s_\infty}{\omega_X (1-s_\infty)} \right]^{1/\varphi}. \quad (106)$$

Define $f_\infty = f(x_\infty) > 0$. In particular, $y \leq f_\infty k^\alpha$ and $0 < u \leq x_\infty/B$.

The static condition can be written

$$k^\alpha v(x) = 1/B, \quad v(x) = \frac{(1-\alpha)s_X f(x)(1-e_X)}{x}.$$

On $(0, x_\infty)$, v is strictly decreasing, from infinity to zero. Indeed, with a prime here denoting differentiation with respect to $\log x$, $e'_X = (\alpha - \sigma^{-1})\varphi s_X(1-s_X) > 0$ and $-d \log v / d \log x = e_X + e'_X / (1 - e_X) > \alpha s_X$. Consequently x increases with B and k , while

$$\frac{\partial \log y}{\partial \log k} = \alpha + \frac{\alpha(1-\alpha)s_X}{e_X + e'_X / (1 - e_X)} < 1.$$

Thus y/k decreases with k and increases with B . For k in any compact interval bounded away from zero, as $B \rightarrow \infty$,

$$x \rightarrow x_\infty, \quad y \rightarrow f_\infty k^\alpha, \quad u \rightarrow 0$$

uniformly. These are properties of the static map, not assumptions about the equilibrium path. They also imply that, for any prescribed finite return, sufficiently small k and sufficiently large B make r exceed that return, uniformly over those small values of k .

Step 2. Household and developer optimality bound consumption. Proposition 2 bounds k and y above by finite constants $k_{\max}, y_{\max}$. Euler and $r \geq -\delta$ give $\dot{c}/c \geq -(\delta + \rho + \gamma)$. Integrating the resource constraint over $[t, t+1]$ therefore gives

$$c_t \int_0^1 e^{-(\delta+\rho+\gamma)v} dv \leq \int_t^{t+1} c_s ds \leq k_{\max} + y_{\max}.$$

Hence c has a finite uniform upper bound $c_{\max}$.

Write $D_{t,s} = \exp(-\int_t^s r_v dv)$. Euler implies $\int_t^\infty D_{t,s} C_s ds = C_t/d$. Integrating the household budget (4) and using its TVC (6) shows that the discounted sum of wages and distributions has a finite limit. Since wages are nonnegative, the discounted distributions alone have a limit that is either finite or minus infinity. The latter is impossible: the developer can stop research and obtain nonnegative operating profits. By continuation optimality, its remaining value $V_t = \int_t^\infty D_{t,s} \Pi_s ds$ is therefore finite and nonnegative. It follows that

$$\frac{C_t}{d} = K_t + V_t + \int_t^\infty D_{t,s} w_s L_s ds, \quad c_t \geq dk_t, \quad \int_t^\infty D_{t,s} w_s L_s ds \leq \frac{C_t}{d}. \quad (107)$$

Here V_t is an auxiliary continuation value, not an additional traded asset or a change to the household budget.

Step 3. Future research spending is small before taking growth limits. Fix a positive rate a . Step 1 permits a small $k_a > 0$ and a sufficiently late date such that $r - (n + \gamma) \geq a$ whenever $k \leq k_a$. On $[k_a, k_{\max}]$, the wage equation (12) and the static map give a uniform positive lower bound ℓ_a on w/A . Since $r \geq -\delta$, there is a finite constant $H_a > 0$ such that

$$a \leq r - (n + \gamma) + H_a w/A$$

at every sufficiently late date. Multiplication by $D_{t,s} A_s L_s$, integration, and (107) give

$$\int_t^\infty D_{t,s} A_s L_s ds \leq \frac{A_t L_t + H_a C_t/d}{a} \leq J A_t L_t \quad (108)$$

for a fixed finite J . The omitted terminal term is nonpositive; no discount-rate limit is used.

The research condition (22) and costate equation (23) imply

$$\frac{d(qB)}{dt} = r q B - U + \frac{1-\eta}{\eta} M. \quad (109)$$

Since B is nondecreasing, $U_s \leq x_\infty A_s L_s / B_t$ for $s \geq t$. Integrating (109) first over finite intervals and then using the developer TVC (25) gives

$$q_t B_t + \frac{1-\eta}{\eta} \int_t^\infty D_{t,s} M_s ds = \int_t^\infty D_{t,s} U_s ds \leq \frac{x_\infty J}{B_t} A_t L_t. \quad (110)$$

The right side is finite by (108); nonnegativity then ensures that the research integral is finite as well. Its value divided by $A_t L_t$ tends to zero. This argument requires $0 < \eta < 1$, but neither $\eta \leq 1/2$ nor $\eta < \alpha$.

Step 4. Capital cannot repeatedly approach zero. Let $F(k) = f_\infty k^\alpha - \nu k$. Choose $\varepsilon > 0$ small enough that F is positive and increasing on $(0, \varepsilon]$. Step 1 also permits a sufficiently late date and $a > 0$ such that $\dot{c}/c = r - \rho - \gamma \geq a$ whenever $k \leq \varepsilon$. For any $j \leq \varepsilon$, a feasible path cannot enter $\{k \leq j, c \geq F(j)\}$: there $\dot{k} \leq F(k) - c \leq 0$, whereas c continues to grow at least at rate a , contradicting its upper bound.

If k stayed below ε forever, consumption would likewise exceed its upper bound. Hence the path eventually reaches ε . Suppose arbitrarily late crossings from ε to $\varepsilon/2$ were possible. Let t_0 be the last crossing of ε before the first subsequent crossing t_1 of $\varepsilon/2$. Throughout this interval $k \in [\varepsilon/2, \varepsilon]$ and c increases. Equation (107) gives $c_{t_0} \geq d\varepsilon$, so

$$t_1 - t_0 \leq T_\varepsilon := a^{-1} \log\left(\frac{c_{\max}}{d\varepsilon}\right) < \infty.$$

At t_1 , feasibility requires $c_{t_1} < F(\varepsilon/2)$ by the preceding invariant-region argument. Thus $c_s < F(\varepsilon/2)$ throughout the crossing. Uniform static convergence on the crossing interval yields numbers $\epsilon_{t_0} \rightarrow 0$ such that

$$\dot{k}_s = y_s - u_s - \nu k_s - c_s - m_s \geq -\epsilon_{t_0} - m_s.$$

On this interval r is bounded above by a fixed positive R_ε . Since $n + \gamma > 0$, equation (110) then implies

$$\frac{\varepsilon}{2} \leq T_\varepsilon \epsilon_{t_0} + \int_{t_0}^{t_1} m_s ds \leq T_\varepsilon \epsilon_{t_0} + e^{R_\varepsilon T_\varepsilon} \frac{\eta x_\infty J}{(1-\eta)B_{t_0}} \rightarrow 0,$$

a contradiction. Therefore k is eventually bounded away from zero, and so is $c \geq dk$. In particular, r is now bounded above.

Step 5. Research vanishes pointwise and Ramsey dynamics determine the limit. Differentiating the research condition and using the costate equation gives

$$(1-\eta)g_M = r - \frac{\eta \chi U B^{\eta-1}}{M^{1-\eta}} \leq r. \quad (111)$$

Choose a positive R above all eventual values of r and put $H = R/(1-\eta)$. Then $M_s \geq M_t e^{-H(t-s)}$ on $[t-1, t]$ at sufficiently late dates. Consequently

$$\frac{1}{A_{t-1} L_{t-1}} \int_{t-1}^t D_{t-1,s} M_s ds \geq e^{n+\gamma-R-H} m_t.$$

The left side tends to zero by (110). Hence $m \rightarrow 0$, and Step 1 already gives $u \rightarrow 0$.

The normalized equilibrium equations therefore approach, uniformly on the compact positive region containing the path, the Ramsey system

$$\dot{k} = F(k) - c, \quad \dot{c} = c[\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma]. \quad (112)$$

For clarity, the following argument verifies convergence rather than inferring it solely from convergence of the equations. This limiting system has one positive stationary point,

$$k_\infty = \left[\frac{\alpha f_\infty}{\rho + \gamma + \delta} \right]^{1/(1-\alpha)}, \quad c_\infty = F(k_\infty) > 0.$$

Its Jacobian has negative determinant, so it is a saddle. The standard Ramsey phase portrait is discussed by Groth (2013). There is no nonconstant complete orbit confined to a compact subset of the positive quadrant. To see this, the multiplier $1/c^2$ gives

$$\frac{\partial}{\partial k} \frac{F(k) - c}{c^2} + \frac{\partial}{\partial c} \frac{\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma}{c} = \frac{\rho - n}{c^2} > 0.$$

The Bendixson–Dulac argument excludes both periodic orbits and homoclinic loops. The planar limit-set theorem, the unique stationary point, and its saddle character then exclude a nonconstant complete orbit confined to such a compact set: its forward and backward limits would otherwise require a periodic orbit or a connection back to that same saddle.

For any sequence of dates tending to infinity, time translations of our bounded positive path have a subsequence converging on every bounded time interval to a complete orbit of (112). This follows from bounded derivatives, uniform convergence of the vector fields, and the integral form of the differential equations. The preceding paragraph forces that orbit to be the stationary point. Thus every accumulation point is (k_∞, c_∞) , proving convergence of the original path.

It follows that $y \rightarrow y_\infty = f_\infty k_\infty^\alpha > 0$ and

$$r \rightarrow \rho + \gamma, \quad \frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta}, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0. \quad (113)$$

Positivity follows because the numerator of the final expression is $\rho - n + (1 - \alpha)(\delta + n + \gamma) > 0$. This proves (46). It also gives $U/Y, M/Y \rightarrow 0$, $\dot{k} \rightarrow 0$, and $g_C \rightarrow n + \gamma$.

Step 6. Output and wage growth converge as well. A convergent level need not have a convergent derivative. The bound on g_M from Step 5 and the research law give

$$B_t^{1-\eta} \geq (1 - \eta)\chi M_t^\eta \int_0^1 e^{-\eta H v} dv.$$

Thus $g_B = \chi M^\eta / B^{1-\eta}$ is bounded above. At x_∞ , the marginal-revenue function v defined in Step 1 has a simple zero: $v(x_\infty) = 0$ and $v'(x_\infty) < 0$, where the prime now denotes differentiation with

respect to x . Differentiating $k^\alpha v(x) = 1/B$ yields

$$\dot{x} = -\frac{v(x)}{v'(x)} \left(g_B + \alpha \frac{\dot{k}}{k} \right) \rightarrow 0.$$

Hence $\dot{y} \rightarrow 0$ and $\dot{s}_X \rightarrow 0$. Since $y_\infty > 0$ and $1 - s_\infty > 0$, $g_{Y/N} \rightarrow \gamma$ and, by (12), $g_w \rightarrow \gamma$. The limiting income shares follow by substituting s_∞ into (12) and (13). The result is conditional on equilibrium existence and $B \rightarrow \infty$, not an equilibrium-existence theorem or an assumption of balanced growth. $\square$

Proof of Proposition 4. Use the definitions in Appendix C. First, all candidate ratios are positive. The research denominator $\rho - n + \eta g_Y^*$ is positive. Moreover,

$$i^* = \frac{\alpha(g_Y^* + \delta)}{\rho - n + g_Y^* + \delta} < \alpha, \quad m^* < \frac{u^* \eta}{1 - \eta}.$$

Since $\Delta = (1 - \alpha)(1 - \eta - \omega_X) > 0$, $\omega_X < 1 - \eta$ and $\omega_X < 1$. Consequently,

$$u^* + m^* < \frac{u^*}{1 - \eta} < (1 - \alpha)\omega_X < 1 - \alpha, \quad c^* = 1 - u^* - i^* - m^* > 0.$$

Thus positive consumption follows from the stated conditions; it is not an additional assumption.

For given $A_0, N_0 > 0$, choose the reference initial stocks as

$$B_0^* = \left[A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^*)^{\omega_X/\omega_L} m^* \left(\frac{\chi}{g_B^*} \right)^{1/\eta} \right]^{\eta\omega_L/(1-\eta-\omega_X)}, \quad (114)$$

$$Y_0^* = A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^* B_0^*)^{\omega_X/\omega_L}, \quad K_0^* = k^* Y_0^*. \quad (115)$$

The exponent is positive. These choices solve simultaneously the initial production equation and $g_B^* B_0^* = \chi(B_0^* m^* Y_0^*)^\eta$. Set $U_0^* = u^* Y_0^*$, $M_0^* = m^* Y_0^*$, $C_0^* = c^* Y_0^*$, $X_0^* = B_0^* U_0^*$, and

$$q_0^* = \frac{M_0^*}{\eta g_B^* B_0^*}. \quad (116)$$

Let $Y_t^* = Y_0^* e^{g_Y^* t}$, $B_t^* = B_0^* e^{g_B^* t}$, $K_t^* = k^* Y_t^*$, $C_t^* = c^* Y_t^*$, $U_t^* = u^* Y_t^*$, $M_t^* = m^* Y_t^*$, and $q_t^* = q_0^* e^{(g_Y^* - g_B^*) t}$. Use the exogenous A_t, N_t and set $L_t = N_t$ and $X_t^* = B_t^* U_t^*$. The static equilibrium equations determine all remaining quantities and prices.

The two growth identities (51) and (52) make production and research accumulation hold at every date once they hold initially. The static monopoly condition gives $p_X X = (1 - \alpha)\omega_X Y$ and $U = u^* Y$. Equation (116) gives $M = \eta q \dot{B}$, which is the research first-order condition (28k) with $\psi = 1$. The definitions of k^* , r^* , and c^* verify the capital-return, Euler, and resource equations.

The costate equation reduces to

$$g_Y^* - g_B^* = r^* - \frac{u^* \eta g_B^*}{m^*} - \eta g_B^*.$$

This holds by equation (136) and $(1 - \eta)g_B^* = \eta g_Y^*$. All allocations are positive and finite at every finite date.

Both transversality conditions are explicit:

$$e^{-\rho t} \frac{N_t K_t^*}{C_t^*} = \frac{N_0 k^*}{c^*} e^{-(\rho-n)t} \longrightarrow 0, \quad (117)$$

$$e^{-r^* t} q_t^* B_t^* = q_0^* B_0^* e^{-(\rho-n)t} \longrightarrow 0. \quad (118)$$

Discounted developer profit is a constant multiple of $e^{-(\rho-n)t}$. Discounted household utility is $N_0 e^{-(\rho-n)t} [\log(C_0^*/N_0) + (g_Y^* - n)t]$. Both improper objective integrals are finite.

It remains to establish global optimality, rather than only the dated necessary conditions. At given paths of K, A, L , static optimization gives

$$\begin{aligned} \pi_t(B) &= (1 - \alpha)\omega_X [1 - (1 - \alpha)\omega_X] [K_t^\alpha (A_t L_t)^{(1-\alpha)\omega_L}]^{1/[1-(1-\alpha)\omega_X]} \\ &\quad \times [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X/[1-(1-\alpha)\omega_X]}. \end{aligned} \quad (119)$$

First suppose $\eta \leq 1/2$. This function is concave in B when $(1 - \alpha)\omega_X \leq 1/2$. The research technology $F(B, M) = \chi B^\eta M^\eta$ is jointly concave when $\eta \leq 1/2$. Since $q > 0$, the developer's current-value Hamiltonian $\pi_t(B) - M + q_t F(B, M)$ is jointly concave, and its first-order conditions maximize it globally. Write $D_t = \exp(-\int_0^t r_s ds)$ and $J_T = \int_0^T D_t [\pi_t(B_t) - M_t] dt$. For any admissible alternative with the same initial B , the supporting hyperplane inequality and the costate equation imply

$$\tilde{J}_T - J_T \leq -D_T q_T (\tilde{B}_T - B_T) \leq D_T q_T B_T \longrightarrow 0. \quad (120)$$

Hence no admissible alternative has a greater limiting payoff, or a greater upper limit of truncated payoff, than this finite-valued candidate. For $\eta > 1/2$, use only within this proof the invertible state transformation and its current-value costate

$$S = B^{1-\eta}, \quad \dot{S} = (1 - \eta)\chi M^\eta, \quad p = \frac{q B^\eta}{1 - \eta}. \quad (121)$$

This is the same feasible research problem, not a different technology. At given K, A, L , equation (119) becomes a positive time-dependent coefficient times

$$S^{\frac{(1-\alpha)\omega_X}{[1-(1-\alpha)\omega_X](1-\eta)}}.$$

It is concave in S if $(1 - \alpha)\omega_X \leq (1 - \eta)/(2 - \eta)$. The maintained restrictions imply

$$(1 - \alpha)\omega_X < (1 - \eta)^2 < \frac{1 - \eta}{2 - \eta} \quad (\eta > 1/2).$$

Consequently $\pi_t(S^{1/(1-\eta)} - M + p_t(1-\eta)\chi M^\eta)$ is jointly concave in S, M . The transformed first-order and costate conditions follow by the chain rule from the original equations. Moreover, $pS = qB/(1 - \eta)$, so the original developer TVC implies $D_T p_T S_T \rightarrow 0$. Applying the supporting-hyperplane argument in these coordinates yields

$$\tilde{J}_T - J_T \leq -D_T p_T (\tilde{S}_T - S_T) \leq \frac{D_T q_T B_T}{1 - \eta} \rightarrow 0.$$

Thus both ranges of η give global developer optimality against all admissible research paths. The argument applies to any candidate satisfying the uncapped equations and developer TVC under the proposition's restrictions, not just the BGP.

For the household, set $\Lambda_t = e^{-\rho t} N_t / C_t$. Euler implies $\dot{\Lambda} = -r\Lambda$. Concavity of log utility and the linear budget give, for any feasible alternative with the same K_0 and the same given prices and distributions,

$$\int_0^T e^{-\rho t} N_t \left[\log(\tilde{C}_t / N_t) - \log(C_t / N_t) \right] dt \leq -\Lambda_T (\tilde{K}_T - K_T) \leq \Lambda_T K_T \rightarrow 0.$$

This establishes global household optimality. Competitive final-good firms face a concave constant-returns technology, and Lemma 1 establishes global static monopoly optimality. The constructed path satisfies Definition 1 with the uncapped research law (42). $\square$

Proof of Proposition 6. Use the ordering $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$ in (138), and let e_K, e_B, e_C, e_q be the corresponding unit row vectors. Define the row vectors

$$y_\xi = \left(\frac{\alpha}{1 - (1 - \alpha)\omega_X}, \frac{(1 - \alpha)\omega_X}{1 - (1 - \alpha)\omega_X}, 0, 0 \right), \quad (122)$$

$$m_\xi = \frac{\eta e_B + e_q}{1 - \eta}, \quad (123)$$

$$b_\xi = g_B^*[(\eta - 1)e_B + \eta m_\xi]. \quad (124)$$

The reduced production equation gives the log deviation of output as $y_\xi \xi$. The research FOC implies $M = (q\chi\eta B^\eta)^{1/(1-\eta)}$, so the log deviation of M is $m_\xi \xi$. Differentiating $g_B = \chi B^{\eta-1} M^\eta$ at the BGP gives the level deviation $b_\xi \xi$.

The four rows of J^* , derived from capital accumulation, AI research, the household Euler

equation, and the developer costate equation, are

$$J_K^* = \frac{1 - u^*}{k^*} y_\xi - (g_Y^* + \delta) e_K - \frac{c^*}{k^*} e_C - \frac{m^*}{k^*} m_\xi, \quad (125)$$

$$J_B^* = b_\xi, \quad (126)$$

$$J_C^* = (r^* + \delta)(y_\xi - e_K), \quad (127)$$

$$J_q^* = (r^* + \delta)(y_\xi - e_K) - (\rho - n + \eta g_Y^*)(y_\xi - e_q - e_B) - \eta b_\xi. \quad (128)$$

The last row uses $U/(qB) = \rho - n + \eta g_Y^*$ at the BGP. All entries depend on growth rates and ratios, which are independent of χ . No terminal restrictions were used to derive this Jacobian.

The exact normalized vector field is continuously differentiable near $\xi = 0$. By the assumed hyperbolicity and the two stable eigenvalues, the stable-manifold theorem supplies a two-dimensional local manifold whose trajectories remain near zero and converge exponentially. The derivative of its projection onto (ξ_K, ξ_B) is V_{KB} . The inverse-function theorem therefore makes this manifold locally the graph of a continuously differentiable function selecting (ξ_C, ξ_q) from the two initial stock deviations. Since exponentiation is a local diffeomorphism, the same holds for (C_0, q_0) as functions of (K_0, B_0) . Shrinking the neighborhood preserves positivity, feasibility, and existence for every $t \geq 0$. This is local uniqueness among trajectories that remain in the neighborhood and converge to the BGP.

Exponential convergence implies that $r_t - r^*$ is integrable. Thus D_t/e^{-r^*t} tends to a finite positive constant, and both TVC expressions equal their BGP counterparts in (117) and (118) times factors tending to finite positive limits. Both TVCs hold. Output and all expenditure flows are bounded multiples of $e^{g_Y^*t}$, while log consumption per person is a linear function of time plus a bounded deviation. Both objective integrals therefore remain finite. The global concavity verification in the proof of Proposition 4 applies at these new price and allocation paths, proving agent optimality. Static optimality and market clearing then establish equilibrium, not merely a solution of the necessary conditions. $\square$

B Numerical algorithm and equilibrium audit

B.1 Economic intuition and algorithm in words

The numerical problem is to select an equilibrium path, not merely to integrate the laws of motion from four arbitrary initial values. Capital and AI efficiency are predetermined at date zero, whereas consumption and the developer's shadow value can jump. For given K_0 and B_0 , almost every choice of C_0 and q_0 sends the economy away from the terminal equilibrium regime or violates feasibility. The algorithm therefore chooses C_0 and q_0 jointly with the entire path so that the economy starts from the prescribed stocks and approaches the appropriate stable continuation.

The calculation proceeds in six steps:

1. For each value of σ , I use the analytical results in Section 4 to identify the relevant terminal

regime and its normalized limiting point.

2. Near that point, I construct a local boundary-value solution. The terminal restrictions eliminate the two unstable directions rather than fixing arbitrary terminal levels for the jump variables.
3. Conditional on every trial value of the dynamic variables, I solve the static monopoly condition for AI production services and recover research expenditure from the developer's first-order condition. Thus the numerical system uses the static choices already derived in Section 3; it does not add behavioral equations.
4. I use continuation to move the local solution gradually to the common prescribed values of K_0 , B_0 , and $A_0 N_0$. Each successful solution is the initial guess for the next continuation stage. This step selects the two date-zero jump variables without treating the no-AI economy as a mechanical starting point.
5. I lengthen the terminal horizon and tighten the solver tolerances. A valid approximation should leave the date-zero jumps and the entire displayed path essentially unchanged.
6. I admit the path only after separate checks of feasibility, the original equilibrium equations, static optimality, global developer optimality, both transversality conditions, and consistency with the analytical terminal continuation. A converged boundary-value calculation that fails any of these checks is not reported as an equilibrium.

This procedure can be viewed as finding the stable saddle path from both ends. The initial boundary supplies the predetermined stocks, while the terminal boundary rules out explosive components. Collocation chooses the whole path at once and, with it, the initial consumption and shadow value that reconcile the two boundaries. The terminal date is a computational truncation; the infinite-horizon interpretation comes from the analytical continuation and the transversality checks described below.

B.2 Notation in the paper and code

The numerical code retains its existing variable names. To make the implementation auditable, the notation used in the paper maps into the code as follows:

$$\gamma = \text{labor_productivity_growth}, \quad \sigma = \text{sigma_xl}, \quad \psi(B) = \exp(\log_psi). \quad (129)$$

This map changes only the notation in the exposition. It does not redefine a variable or parameter, alter its timing, or modify any equation used by the simulations.

B.3 Boundary-value formulation

A boundary-value problem (BVP) selects the initial consumption and shadow price jointly with the subsequent path. I use the existing collocation solver, `scipy.integrate.solve_bvp`, for the four

dynamic equations in the equilibrium definition. At every evaluation, I solve the static monopoly condition and recover research expenditure from its first-order condition. Effective labor is exogenous, $AL = A_0 N_0 e^{(n+\gamma)t}$, and therefore is not an additional unknown path in this dated BVP.

The computational coordinates are

$$\left(\log K, \log \frac{B}{\bar{B} - B}, \log C, \log q \right).$$

They enforce positivity and $0 < B < \bar{B}$ without clipping a computed equilibrium. I subtract the analytical terminal aggregate growth rate times t from each coordinate to remove its linear trend. This is a numerical transformation, not a change in the model's variables. In particular, the code stores $\log \psi(B)$ separately: a displayed ratio $B/\bar{B}$ can round to one even though the remaining gap to the upper bound is strictly positive.

The left boundary fixes only K_0 and B_0 : from (60) in the main comparison and from (65) in the Ramsey-start experiment, (66)–(67) in the near-terminal diagnostic, and (68) in the slow-transition exercise. At the right boundary, the solution is mapped into the regime-specific normalized coordinates of Appendix A. Two projection conditions remove the unstable components of the linearized terminal system. Its third stable direction is associated with the exogenous scale. The two stock conditions and two terminal projections thus close a four-dimensional problem; neither C_0 nor q_0 is predetermined. A finite terminal projection approximates the stable continuation, which is why increasing the horizon is a necessary numerical check.

For $\sigma \leq 1$, I use the positive-labor-share terminal formulas. For $\sigma > 1$, the upper bound's position relative to $\mathcal{B}(\sigma)$ selects the terminal regime. At $\sigma = 1$, the CES block uses its exact Cobb–Douglas limit. Near one, `log1p` and `expm1` evaluate the CES expression without cancellation, and the comparison between the upper bound and the threshold is performed in logarithms. I solve the labor-supported terminal static condition directly for $\log[X/(AL)]$, rather than invert a share formula by dividing by $\sigma - 1$. A trial quantity with nonpositive marginal revenue can bracket the monopoly optimum, but is never accepted as that optimum.

B.4 Continuation and numerical accuracy

For each elasticity, I first solve a local BVP near its analytical terminal point. I then move the initial stocks and effective-labor scale continuously to their prescribed values, retaining the preceding solution as the next guess. Each design uses 32 continuation stages. There is no continuation through the degenerate no-AI economy. The common main stocks are those in (60); the uncapped unit-elastic reference supplies stocks for the slow-transition and financing sensitivities, but never supplies their jump variables.

The initial horizon combines the time needed to reach the terminal scale with eleven decay times of the slowest stable eigenvalue. This gives an initial truncation criterion, not an equilibrium assumption. I extend that horizon twice by 500 years, tightening the collocation tolerance from 2×10^{-6} to 10^{-8} and then 10^{-9} . The final boundary tolerance is 10^{-11} . Initial meshes of 221,

401, and 601 points are refined adaptively by the solver. The final horizons exceed the 500-year display windows and the slow exercise’s 4,000-year window, as applicable; the figures contain no extrapolated spline values.

Table 4: Numerical accuracy of the four displayed equilibria

σ	Final horizon	Mesh points	Dynamic residual	Horizon change
0.90	4,946.2	874	9.17×10^{-10}	3.26×10^{-8}
1.00	4,821.1	872	1.05×10^{-9}	3.73×10^{-8}
1.10	4,737.0	909	9.12×10^{-10}	3.37×10^{-8}
1.50	3,008.5	1,092	8.19×10^{-10}	1.56×10^{-8}

Dynamic residuals reconstruct the four original equations using five-point differences of spline values at 1,001 dates, with steps of 0.003 and 0.001 years; the table reports the larger maximum. Horizon change is the largest change in the detrended logarithmic coordinates over years 0–500 between the last two solves. Values are rounded upward.

The slower-transition exercise repeats this entire calculation rather than rescaling or extending the main trajectories. Its final horizons are 6,857.1, 6,731.9, 6,647.9, and 4,919.4 years for $\sigma = 0.90, 1, 1.10, 1.50$, respectively. The corresponding independently reconstructed dynamic residuals are below 1.1×10^{-9} , and extending the horizon changes the detrended coordinates by at most 2.8×10^{-8} over the full 0–4,000 display window.

The Ramsey-start experiment also repeats the complete calculation for four separate BVPs. Its final horizons are 4,946.2, 4,821.1, 4,737.0, and 3,008.5 years for $\sigma = 0.90, 1, 1.10, 1.50$. The independently reconstructed dynamic residuals are below 1.1×10^{-9} , and the final horizon extension changes the detrended coordinates by at most 3.8×10^{-8} over years 0–500.

The maximum research and static monopoly first-order residuals are below 1.8×10^{-13} . Initial jump choices change by less than 1.8×10^{-10} in logs in the final horizon comparison. The normalized terminal coordinates differ from their analytical limits by less than 2.3×10^{-6} . The acceptance thresholds are deliberately looser than these observed errors: 10^{-6} for independently reconstructed dynamic residuals, 10^{-9} for static first-order conditions, 2×10^{-5} for horizon sensitivity, and 10^{-4} for terminal-coordinate gaps. Tightening the solver and extending its horizon provide the evidence of stability; the thresholds do not replace any economic condition.

B.5 Optimality and the infinite-horizon continuation

The equation audit also checks feasibility, the monopoly second-order condition, and that numerical safeguards used for off-path Newton trials do not bind on the solution. Both transversality expressions converge to zero on each selected continuation: their asymptotic log growth rate is $n - \rho = -0.037$. This analytical rate, rather than a small terminal value alone, establishes the transversality check.

For $\sigma = 0.90, 1, 1.10$, the developer’s operating profit is concave in research depth over the counterfactual AI-efficiency domain checked at each date. The final audit uses 161 dates and 161 AI-efficiency points per date, with scalar minimization between grid points. For $\sigma = 1.50$, this

stronger curvature test fails at low counterfactual AI-efficiency levels; checking concavity only along the realized path would miss that failure.

I instead apply the support condition in Lemma 2. At each date, I minimize its tangent gap, divided by the candidate’s Y , over counterfactual AI-efficiency levels $B \geq B_0$, refining every local minimum found on the grid. An analytical upper bound on the Hamiltonian’s derivative covers the unbounded research-depth domain beyond the numerical grid. The check is repeated with 81 dates and 101 AI-efficiency points, then with 321 dates and 241 points. In the main, Ramsey-start, and slow-transition comparisons, the smallest normalized gaps are of order -10^{-14} , consistent with floating-point error at the tangent point. The numerical gap tolerance is 10^{-10} ; the economic support inequality remains exact.

For the infinite-horizon continuation, I use (82) with $B_* = 0.75\bar{B}$. Its limiting margin divided by output is approximately $0.09781 > 0$; the computed terminal margin differs from that limit by less than 6×10^{-12} . Thus the analytical continuation satisfies the sufficient support condition eventually, while the dated audit checks the computed transition. These are numerically verified equilibrium approximations supported by analytical terminal and optimality results, not interval-arithmetic existence certificates for the common initial stocks. Nor do they establish existence for every initial condition or every parameter vector.

The financing sensitivity reported in Section 6.7 is solved separately rather than spliced onto a main trajectory. It uses $K_0 = 2.027733654$, $B_0 = 0.443670932$, and $\chi = 1.4223$, preserving the 50-year transition midpoint while changing the predetermined stocks. The calculation repeats both horizon extensions, the independent equation and TVC checks, and the two Hamiltonian-support grids. It passes the same admission thresholds as the main $\sigma = 1.50$ path.

The slow-transition comparison in Section 6.6 likewise uses separate BVPs, checkpoints, audit files, and manifests. It sets $\chi = 0.01$ and uses the earlier stocks in (68); all four paths pass the equation, horizon, TVC, and developer-optimality gates before export.

The Ramsey-start comparison in Section 6.4 begins directly on the positive-AI branch with the predetermined stocks in (65). The numerical continuation changes those initial stocks, not ω_X : it never passes through the degenerate no-AI problem. All four paths pass the same equation, horizon, TVC, and developer-optimality gates before export.

B.6 Reproduction and export safeguards

The implementation preserves the earlier solver. The entry point is `scripts/simulate_rewrite_finite_frontier.py`; the final audit is `scripts/audit_rewrite_equilibria.py`. The reproducible sequence is: solve each of the four elasticities with `--sigma`; run `--verify-long-horizon`; run `audit_rewrite_hamiltonian_support.py` for $\sigma = 1.50$ at both grid resolutions; run the final audit; run `audit_initial_financing_sensitivity.py`; export; and render with `plot_rewrite_equilibria.py`. The public reproduction command performs that sequence for the main, Ramsey-start, slow, and near-terminal designs, using display horizons of 500 years except for the 4,000-year slow exercise. All code, audit files, plot-ready data, and figures used in these simulations are

available in the paper’s [public GitHub repository](#). The repository’s [replication guide](#) provides a one-command workflow, installation instructions, an output map, and the criteria used to interpret the diagnostics. The execution record `audit/rewrite_equilibrium_simulations.md` reports the commands and test results for the version used in the paper.

The exported data and figures require all four scenarios to pass admission. Checkpoint hashes link each audit to its solved spline; a second manifest links the plotted CSV to those checkpoints. Failed or stale candidates cannot silently enter a partial comparison. The data retain the original model variables and use only the normalizations stated in the figure captions.

C The uncapped economy with unit elasticity

This appendix derives the rates and allocation ratios supporting Proposition 4 in Section 5.2 and establishes local equilibrium trajectories near its BGP. I use the uncapped research law (42) and the production function (50). In this appendix and its proofs, a superscript $*$ denotes the reference BGP characterized in Section 5.2, distinguishing it from nearby equilibrium trajectories.

C.1 An explicit balanced-growth equilibrium

At unit elasticity, $s_X = \omega_X$ and $e_X = 1 - (1 - \alpha)\omega_X$. The AI-demand and monopoly conditions, equations (13) and (19), give

$$\frac{p_X X}{Y} = (1 - \alpha)\omega_X, \quad \frac{U}{Y} = (1 - \alpha)^2 \omega_X^2. \quad (130)$$

Substituting $X = BU = (1 - \alpha)^2 \omega_X^2 BY$ into production yields

$$Y^{1-(1-\alpha)\omega_X} = K^\alpha (AL)^{(1-\alpha)\omega_L} [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X}. \quad (131)$$

On an exact balanced-growth path with positive constant K/Y , C/Y , and M/Y , log differentiation of (131) and (42) gives the two growth restrictions (51) and (52). Define

$$\Delta \equiv (1 - \alpha)(1 - \eta - \omega_X). \quad (132)$$

For $\Delta > 0$, their unique solution has

$$g_Y^* = \frac{(1 - \eta)(1 - \omega_X)}{1 - \eta - \omega_X} (n + \gamma), \quad (133)$$

and g_B^* as in (57). The equality $g_K = g_C = g_Y^*$ follows from constant capital and consumption ratios; equations (12) and (5) then give (54) and (55).

Define the candidate ratios $k^* = K/Y$, $u^* = U/Y$, $i^* = (\dot{K} + \delta K)/Y$, $m^* = M/Y$, and

$c^* = C/Y$ by

$$k^* = \frac{\alpha}{r^* + \delta}, \quad u^* = (1 - \alpha)^2 \omega_X^2, \quad (134)$$

$$i^* = (g_Y^* + \delta)k^*, \quad (135)$$

$$m^* = \frac{u^* \eta g_B^*}{\rho - n + \eta g_Y^*}, \quad (136)$$

$$c^* = 1 - u^* - i^* - m^*. \quad (137)$$

In this appendix, k^* is a capital–output ratio, not capital per unit of effective labor. The research share comes from the developer’s first-order and costate conditions; the consumption share comes from market clearing. These ratios are endogenous, not fixed saving or research rules. Along the BGP, $B_t = B_0^* e^{g_B^* t} \rightarrow \infty$, $g_K = g_C = g_U = g_M = g_Y^*$, $g_X = g_Y^* + g_B^*$, and $g_q = g_Y^* - g_B^*$.

The conditions have different roles. Since $n + \gamma > 0$, equations (51) and (52) require $\Delta > 0$ for this positive finite-rate BGP. If $\Delta = 0$, they are inconsistent; if $\Delta < 0$, they imply $g_B < 0$, contrary to the research law. Thus $\eta + \omega_X \geq 1$ excludes this BGP, not every possible equilibrium trajectory. When $\eta \leq 1/2$, the bound $(1 - \alpha)\omega_X \leq 1/2$ makes operating profit concave in B , and research technology is jointly concave in B, M . When $\eta > 1/2$, the proof instead uses the state $S = B^{1-\eta}$. The restrictions $\eta < \alpha$ and $\eta + \omega_X < 1$ then ensure concavity in the transformed problem. Thus $\eta < \alpha$ supports this second verification argument, although the BGP growth identities do not require it separately. None of these concavity tests is asserted to be necessary for equilibrium existence. Finally, $\rho > n$ makes both transversality expressions decline and the objective integrals converge.

C.2 Equilibrium trajectories near the balanced-growth path

The BGP supplies a reference for proving existence from nearby initial stocks that need not be on that path. Define stationary log deviations

$$\xi_K = \log \frac{K}{K^*}, \quad \xi_B = \log \frac{B}{B^*}, \quad \xi_C = \log \frac{C}{C^*}, \quad \xi_q = \log \frac{q}{q^*}. \quad (138)$$

Substitution into the four equilibrium dynamics gives a smooth autonomous system $\dot{\xi} = f(\xi)$ with $f(0) = 0$, where $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$. Let $J^* = Df(0)$ denote its Jacobian. The proof in Appendix A derives its entries directly from equations (29a)–(29d), with $\psi = 1$ and $\psi' = 0$.

Proposition 6 (Local existence of uncapped equilibrium trajectories). *Under the assumptions of Proposition 4, fix $A_0, N_0 > 0$ and their reference BGP. Suppose J^* has no eigenvalue with zero real part and exactly two with negative real parts. Let $V_S \in \mathbb{R}^{4 \times 2}$ be a real basis of its stable eigenspace and V_{KB} its first two rows. If $\det V_{KB} \neq 0$, then every (K_0, B_0) in a sufficiently small open neighborhood of (K_0^*, B_0^*) has an infinite-horizon equilibrium trajectory converging to the BGP in the normalized variables (138). The required (C_0, q_0) are continuously differentiable functions of (K_0, B_0) . The trajectory satisfies both transversality conditions and global agent optimality.*

Uniqueness is local, among paths that remain in the specified neighborhood and converge to the reference BGP.

Proof. See Appendix A, [Proof of Proposition 6](#). □

No constant growth rates or allocation shares are imposed along these transitions. The result is local in initial stocks, not in time; it does not establish existence or uniqueness from arbitrary initial conditions. For the parameters in Table 2, the eigenvalues are approximately -0.098971 , -0.003194 , 0.040391 , and 0.149290 . The absolute determinant of the state projection of unit-length stable eigenvectors is 0.6037 . The two spectral conditions therefore hold at this calibration. The normalized Jacobian is independent of χ ; changing research productivity changes the reference levels, not these roots.

C.3 Growth and distribution without an upper bound

The uncapped BGP has

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad \frac{(r + \delta)K}{Y} = \alpha, \quad \frac{\Pi}{Y} = (1 - \alpha)\omega_X - u^* - m^*. \quad (139)$$

Equation (54) shows that real wages and output per person grow together at a rate above γ , even though labor retains a positive, constant income share. Without an upper bound, continued growth in AI efficiency sustains this wage-growth premium. With any finite upper bound, the unit-elastic equilibria in Proposition 1(ii) instead have $g_{Y/N}, g_w \rightarrow \gamma$. The equivalence between wage growth above γ and a vanishing labor share across the regimes with a finite upper bound does not extend to the uncapped economy.

Within the restrictions of Proposition 4, increasing the AI weight raises the common growth rate and the interest rate:

$$\frac{\partial g_{Y/N}^*}{\partial \omega_X} = \frac{\partial g_w^*}{\partial \omega_X} = \frac{\partial r^*}{\partial \omega_X} = \frac{\eta(1 - \eta)(n + \gamma)}{(1 - \eta - \omega_X)^2} > 0. \quad (140)$$

Labor's income share falls across these parameterizations, but remains constant over time on each BGP. As $\omega_X \rightarrow 0^+$, the common per-person growth rate converges to γ and $r^* \rightarrow \rho + \gamma$. These statements compare balanced-growth equilibria; they do not describe a transition from a no-AI economy or establish an uncapped equilibrium for $\sigma > 1$.

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